

Sieve of Eratosthenes (1–100)

Mathematical description. Fix $n = 100$. Let $S = \{2, 3, \dots, n\}$. For each integer p defined as the smallest element of S not yet processed, declare p *prime* and remove from S every integer of the form kp with $k \geq 2$. Continue this procedure while $p^2 \leq n$. The elements of S that remain after this process are precisely the prime numbers $\leq n$.

Algorithm (short).

1. Initialize the list $2, 3, \dots, n$.
2. For each current smallest unprocessed p (starting at $p = 2$), remove all multiples $2p, 3p, 4p, \dots$ up to n .
3. Stop once $p^2 > n$ (i.e. $p > \lfloor \sqrt{n} \rfloor$). Remaining numbers are primes.

Note. 1 is neither prime nor composite.

	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Primes The primes ≤ 100 are:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

0.1 L1: 1.1 Divisibility (syllabus, expectations)

0.2 L2: 1.2 Prime numbers (open conjectures, breakthroughs)

0.3 L3: 1.3 GCD, 1.4 Euclidean Algorithm

0.4 L4: 1.5 Fundamental Theorem of Arithmetic

0.5 L5: 1.5 Dirichlet's Theorem, 2.1 Congruences

Lecture 5: Dirichlet's Theorem and applications of the FTA

§ Theorem 1.21 (Dirichlet's theorem on primes in arithmetic progressions) Let $a, b \in \mathbb{Z}$ with $b > 0$ and $\gcd(a, b) = 1$. Then the arithmetic progression

$$a, a + b, a + 2b, a + 3b, \dots$$

contains infinitely many prime numbers.

§ Remark (on Theorem ??) Setting $a = 1$ and $b = 1$ recovers the elementary statement that there are infinitely many primes. (We remark that the full proof of Dirichlet's theorem requires analytic tools and is beyond the scope of this course.)

§ Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4) There are infinitely many primes of the form $4n + 3$, where n is a nonnegative integer.

§ Lemma 1.23 (Product of numbers congruent to 1 modulo 4) Let $a, b \in \mathbb{Z}$. If $a \equiv 1 \pmod{4}$ and $b \equiv 1 \pmod{4}$, then $ab \equiv 1 \pmod{4}$.

Proof. Write $a = 4n_1 + 1$ and $b = 4n_2 + 1$ with $n_1, n_2 \in \mathbb{Z}$. Then

$$ab = (4n_1 + 1)(4n_2 + 1) = 16n_1n_2 + 4n_1 + 4n_2 + 1 = 4(4n_1n_2 + n_1 + n_2) + 1,$$

so $ab \equiv 1 \pmod{4}$. This proves the lemma. $\square$

Proof of Proposition ??. Suppose, for contradiction, that there are only finitely many primes congruent to 3 modulo 4. List them as

$$P_1 = 3, P_2, P_3, \dots, P_r,$$

all distinct and each $P_i \equiv 3 \pmod{4}$.

Consider the integer

$$N := 4P_1P_2 \cdots P_r - 1.$$

Note that $N \equiv -1 \equiv 3 \pmod{4}$.

Let q be any prime divisor of N . We make two observations:

1. For each $i = 1, \dots, r$ we have $N \equiv -1 \pmod{P_i}$, so $P_i \nmid N$. Thus q is not equal to any of the P_i .
2. Since $N \equiv 3 \pmod{4}$, it is not possible that every prime divisor of N is congruent to 1 modulo 4. Indeed, a product of integers each congruent to 1 (mod 4) is itself congruent to 1 (mod 4) by Lemma ??, contradicting $N \equiv 3 \pmod{4}$. Hence some prime divisor q of N satisfies $q \equiv 3 \pmod{4}$.

Combining the two observations, q is a prime congruent to 3 modulo 4 that is not among $P_1, \dots, P_r$, contradicting the assumed finiteness of such primes. Therefore there are infinitely many primes congruent to 3 modulo 4. $\square$

§ Remark (on Proposition ??) The method above—building an integer with residue 3 (mod 4) whose prime divisors cannot all be from a given finite list—gives a proof in this special case of Dirichlet's theorem. However, this argument does not extend directly to all arithmetic progressions; for the full theorem one needs deeper tools (Dirichlet characters and L -series).

Congruences

§ Historical remark (Gauss and congruences) Up to the beginning of the nineteenth century number theory consisted of many isolated results. In 1801, Gauss, in his *Disquisitiones Arithmeticae*, introduced the modern theory of congruences and unified many isolated results in number theory.

§ Definition (Congruence modulo m) Let $a, b, m \in \mathbb{Z}$ with $m > 0$. We say that a is congruent to b modulo m , and write

$$a \equiv b \pmod{m},$$

if $m \mid (a - b)$. The integer m is called the *modulus* of the congruence.

§ Remark (notation) The notation $a \not\equiv b \pmod{m}$ means $m \nmid (a - b)$. The modulus is customarily taken to be positive; we will assume $m > 0$ in the sequel.

§ Proposition 2.1 (Congruence modulo m is an equivalence relation) Congruence modulo m is reflexive, symmetric, and transitive on $\mathbb{Z}$; hence it is an equivalence relation.

Proof. Reflexive: For any $a \in \mathbb{Z}$, $a - a = 0$ and $m \mid 0$, so $a \equiv a \pmod{m}$.

Symmetric: If $a \equiv b \pmod{m}$ then $m \mid (a - b)$, hence $m \mid -(a - b) = (b - a)$, so $b \equiv a \pmod{m}$.

Transitive: If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$ then $m \mid (a - b)$ and $m \mid (b - c)$, therefore $m \mid (a - b) + (b - c) = a - c$, so $a \equiv c \pmod{m}$. $\square$

§ Consequence 2.2 (Partition into residue classes) The set $\mathbb{Z}$ is partitioned into equivalence classes under congruence modulo m .

§ Definition (Notation for residue class) For $x \in \mathbb{Z}$, write $[x]_m$ (or simply $[x]$ when m is understood) for the equivalence class of x modulo m :

$$[x]_m = \{y \in \mathbb{Z} : y \equiv x \pmod{m}\}.$$

(Do not confuse $[x]_m$ with the floor function.)

Example: The equivalence classes modulo 4

$$[0]_4 = \{x \in \mathbb{Z} : x \equiv 0 \pmod{4}\} = \{4n : n \in \mathbb{Z}\} = \{\dots, -8, -4, 0, 4, 8, \dots\},$$

$$[1]_4 = \{x : x = 4n + 1, n \in \mathbb{Z}\} = \{\dots, -7, -3, 1, 5, 9, \dots\},$$

$$[2]_4 = \{x : x = 4n + 2, n \in \mathbb{Z}\} = \{\dots, -6, -2, 2, 6, 10, \dots\},$$

$$[3]_4 = \{x : x = 4n + 3, n \in \mathbb{Z}\} = \{\dots, -5, -1, 3, 7, 11, \dots\}.$$

Thus $\mathbb{Z}$ is partitioned into four residue classes modulo 4, and every integer is congruent to exactly one of 0, 1, 2, 3 modulo 4.

§ Definition (Complete residue system modulo m) A set $R \subset \mathbb{Z}$ is a *complete residue system modulo m* if every integer is congruent modulo m to exactly one element of R .

§ Example (complete residue systems) The set $\{0, 1, 2, \dots, m - 1\}$ is a complete residue system modulo m . Also, for example, $\{6, -11, 19, 1988\}$ is a complete residue system modulo 4, since $6 \equiv 2$, $-11 \equiv 1$, $19 \equiv 3$, and $1988 \equiv 0 \pmod{4}$.

§ Proposition 2.3 (Standard complete residue system) The set $\{0, 1, 2, \dots, m - 1\}$ is a complete residue system modulo m .

Proof. Let $a \in \mathbb{Z}$. By the Division Algorithm there exist unique $q, r \in \mathbb{Z}$ with $0 \leq r < m$ such that $a = mq + r$. Then $a - r = mq$, so $m \mid (a - r)$ and $a \equiv r \pmod{m}$, with $r \in \{0, 1, \dots, m - 1\}$. This shows every integer is congruent to at least one element of $\{0, 1, \dots, m - 1\}$.

To show uniqueness: suppose $r_1, r_2 \in \{0, 1, \dots, m-1\}$ and $r_1 \equiv r_2 \pmod{m}$. Then $m \mid (r_1 - r_2)$, but $-(m-1) \leq r_1 - r_2 \leq m-1$. The only multiple of m in this range is 0, so $r_1 - r_2 = 0$ and hence $r_1 = r_2$. Thus each integer is congruent to exactly one element of $\{0, 1, \dots, m-1\}$. $\square$

§ Proposition 2.4 (Arithmetic with congruences) Let $a, b, c, d \in \mathbb{Z}$ and $m > 0$. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

1. $a + c \equiv b + d \pmod{m}$,
2. $ac \equiv bd \pmod{m}$.

Proof. From $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ we have $m \mid (a - b)$ and $m \mid (c - d)$.

For (1): $m \mid (a - b) + (c - d) = (a + c) - (b + d)$, hence $a + c \equiv b + d \pmod{m}$.

For (2): Note $m \mid (a - b)$ implies $m \mid c(a - b)$, and $m \mid (c - d)$ implies $m \mid b(c - d)$. Add these divisibilities to get $m \mid c(a - b) + b(c - d) = ac - bd$, so $ac \equiv bd \pmod{m}$. $\square$

0.6 L6: 2.1 Congruences, 2.2 Linear congruences (1 var.)

L6: Congruences

§ Remark (Proposition ?? in class terms) Recall that ?? from the previous lecture can be expressed in terms of equivalence classes modulo m : for residue classes $[a], [c] \in \mathbb{Z}_m$,

$$[a] + [c] = [a + c] \quad \text{and} \quad [a][c] = [ac].$$

This viewpoint allows construction of addition and multiplication tables for residue classes (modulo m) using any complete residue system.

Examples: tables for $\mathbb{Z}_4$ and $\mathbb{Z}_5$

+	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[0]_4$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[1]_4$	$[1]_4$	$[2]_4$	$[3]_4$	$[0]_4$
$[2]_4$	$[2]_4$	$[3]_4$	$[0]_4$	$[1]_4$
$[3]_4$	$[3]_4$	$[0]_4$	$[1]_4$	$[2]_4$

·	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[0]_4$	$[0]_4$	$[0]_4$	$[0]_4$	$[0]_4$
$[1]_4$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[2]_4$	$[0]_4$	$[2]_4$	$[0]_4$	$[2]_4$
$[3]_4$	$[0]_4$	$[3]_4$	$[2]_4$	$[1]_4$

§ Remark (structure of $\mathbb{Z}_4$) Set $\mathbb{Z}_4 := \{[0]_4, [1]_4, [2]_4, [3]_4\}$. With the above addition and multiplication it is a ring, but not a field: not every nonzero element has a multiplicative inverse (e.g., $[2]_4$ is a zero divisor).

+	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[0]_5$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[1]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$	$[0]_5$
$[2]_5$	$[2]_5$	$[3]_5$	$[4]_5$	$[0]_5$	$[1]_5$
$[3]_5$	$[3]_5$	$[4]_5$	$[0]_5$	$[1]_5$	$[2]_5$
$[4]_5$	$[4]_5$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$

·	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[0]_5$	$[0]_5$	$[0]_5$	$[0]_5$	$[0]_5$	$[0]_5$
$[1]_5$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[2]_5$	$[0]_5$	$[2]_5$	$[4]_5$	$[1]_5$	$[3]_5$
$[3]_5$	$[0]_5$	$[3]_5$	$[1]_5$	$[4]_5$	$[2]_5$
$[4]_5$	$[0]_5$	$[4]_5$	$[3]_5$	$[2]_5$	$[1]_5$

§ Remark (structure of $\mathbb{Z}_5$) Let $\mathbb{Z}_5 := \{[0]_5, [1]_5, [2]_5, [3]_5, [4]_5\}$. Because 5 is prime every nonzero class has a multiplicative inverse, so $\mathbb{Z}_5$ is a field.

§ Question (why extra structure?) Why does $\mathbb{Z}_5$ have more structure (a field) than $\mathbb{Z}_4$ (only a ring)?

§ Answer (primality of modulus) Essentially: $\mathbb{Z}_m$ is a field if and only if m is prime. If m is composite, then some nonzero classes are zero divisors (no inverses). If $m = p$ is prime, then every nonzero class has an inverse (standard proof via Bezout's identity).

Cancellation and a useful proposition

§ Proposition 2.5 (Cancellation in congruences) Let $a, b, c, m \in \mathbb{Z}$ with $m > 0$, and let $d = \gcd(c, m)$. Then

$$ca \equiv cb \pmod{m} \iff a \equiv b \pmod{m/d}.$$

Proof. Suppose $ca \equiv cb \pmod{m}$, i.e. $m \mid c(a - b)$. Let $d = \gcd(c, m)$. Write $c = d \cdot c'$ and $m = d \cdot m'$ so that $\gcd(c', m') = 1$. From $m \mid c(a - b)$ we get $dm' \mid dc'(a - b)$, hence $m' \mid c'(a - b)$. Since $\gcd(c', m') = 1$, it follows that $m' \mid (a - b)$, i.e. $a \equiv b \pmod{m'}$, which is $a \equiv b \pmod{m/d}$.

Conversely, if $a \equiv b \pmod{m/d}$ then $m/d \mid (a - b)$, so $m/d \cdot d = m \mid d(a - b)$. But $d \mid c$, so $c(a - b)$ is a multiple of $d(a - b)$, hence $m \mid c(a - b)$ and $ca \equiv cb \pmod{m}$. This proves the equivalence. $\square$

Linear congruences in one variable

§ Definition (linear congruence) A congruence of the form

$$ax \equiv b \pmod{m},$$

with unknown $x \in \mathbb{Z}$ and given integers a, b, m with $m > 0$, is called a *linear congruence* in the variable x .

§ Examples (small checks)

- $2x \equiv 3 \pmod{4}$ has no solution: checking $x \equiv 0, 1, 2, 3 \pmod{4}$ yields no solution.
- $2x \equiv 3 \pmod{5}$ has the unique solution $x \equiv 4 \pmod{5}$.
- $2x \equiv 4 \pmod{6}$ has two incongruent solutions modulo 6: $x \equiv 2, 5 \pmod{6}$.
- $3x \equiv 9 \pmod{6}$ has three incongruent solutions modulo 6: $x \equiv 1, 3, 5 \pmod{6}$.

§ Theorem 2.6 (Solvability and number of solutions of linear congruences) Let $a, b, m \in \mathbb{Z}$ with $m > 0$ and let $d = \gcd(a, m)$. Then:

1. The congruence $ax \equiv b \pmod{m}$ has a solution iff $d \mid b$.
2. If $d \mid b$ then there are exactly d incongruent solutions modulo m . Concretely, if x_0 is one solution, then all solutions are

$$x = x_0 + \frac{m}{d}n, \quad n \in \mathbb{Z},$$

and the classes $x_0 + \frac{m}{d}n$ for $n = 0, 1, \dots, d-1$ give a complete set of incongruent solutions modulo m .

Proof. We use the standard equivalence between the congruence and a linear Diophantine equation. The congruence $ax \equiv b \pmod{m}$ is equivalent to the existence of an integer y such that

$$ax - my = b. \tag{*}$$

Thus solvability of the congruence is equivalent to solvability of the linear Diophantine equation (*).

(1) If (*) has an integer solution, then $d = \gcd(a, m)$ divides the left-hand side $ax - my$ and hence divides b . Conversely, if $d \mid b$, write $b = de$ for some $e \in \mathbb{Z}$. By Bezout's identity (see ?? or the standard result), there exist integers r, s with $ar + ms = d$. Multiply both sides by e to obtain $are + mse = de = b$. Then $x_0 = re$ and $y_0 = -se$ solve (*), so a solution exists.

(2) Fix any solution x_0 of $ax \equiv b \pmod{m}$. For any integer n consider

$$x = x_0 + \frac{m}{d}n.$$

Then

$$ax = ax_0 + a \frac{m}{d}n = ax_0 + \frac{a}{d}mn \equiv ax_0 \equiv b \pmod{m},$$

so each such x is a solution. Thus there are infinitely many integer solutions, all congruent modulo $\frac{m}{d}$ shifts.

Now suppose x is any solution. Then subtracting the equation for x_0 from the equation for x yields

$$a(x - x_0) \equiv 0 \pmod{m},$$

i.e. $m \mid a(x - x_0)$. Write $a = da'$ and $m = dm'$, with $\gcd(a', m') = 1$. Then $m'd \mid da'(x - x_0)$, so $m' \mid a'(x - x_0)$. Since $\gcd(a', m') = 1$ we deduce $m' \mid (x - x_0)$, hence $x - x_0 = m'n = \frac{m}{d}n$ for some $n \in \mathbb{Z}$. This shows every solution has the stated form. Two such solutions

$$x_0 + \frac{m}{d}n_1 \quad \text{and} \quad x_0 + \frac{m}{d}n_2$$

are congruent modulo m iff

$$\frac{m}{d}(n_1 - n_2) \equiv 0 \pmod{m} \iff m \mid \frac{m}{d}(n_1 - n_2).$$

Dividing both sides by m/d yields $d \mid (n_1 - n_2)$, so $n_1 \equiv n_2 \pmod{d}$. Therefore choosing $n = 0, 1, \dots, d-1$ produces exactly d incongruent solutions modulo m . $\square$

§ Corollary 2.7 (Explicit solution set) Under the hypotheses of Theorem ??, if $d = \gcd(a, m)$ and $d \mid b$, then the d incongruent solutions modulo m are

$$x_0 + \frac{m}{d}n, \quad n = 0, 1, \dots, d-1,$$

where x_0 is any particular solution.

Worked example

§ Example (solve $16x \equiv 8 \pmod{28}$) We solve $16x \equiv 8 \pmod{28}$. Compute $d = \gcd(16, 28)$.

Euclidean algorithm:

$$28 = 1 \cdot 16 + 12, \quad 16 = 1 \cdot 12 + 4, \quad 12 = 3 \cdot 4 + 0,$$

so $d = 4$. Since $4 \mid 8$, the congruence has $d = 4$ incongruent solutions modulo 28 by Theorem ??.

One method: divide the congruence by $d = 4$ to get (working modulo $28/4 = 7$)

$$4x \equiv 2 \pmod{7}.$$

Since $4^{-1} \equiv 2 \pmod{7}$ (because $4 \cdot 2 = 8 \equiv 1 \pmod{7}$), multiply both sides by 2:

$$x \equiv 4 \pmod{7}.$$

Thus the solutions modulo 28 are

$$x \equiv 4 + 7n, \quad n = 0, 1, 2, 3,$$

which gives the incongruent solutions $x \equiv 4, 11, 18, 25 \pmod{28}$.

(Alternatively, one can find an explicit x_0 via the extended Euclidean computation: from $4 = 2 \cdot 16 - 1 \cdot 28$ we get $8 = 4 \cdot 16 - 2 \cdot 28$, so $x_0 = 4$ is a particular solution, yielding the same solution set.)

Multiplicative inverses

§ Definition (multiplicative inverse modulo m) If $a \in \mathbb{Z}$ and there exists $x \in \mathbb{Z}$ with $ax \equiv 1 \pmod{m}$, then x is called a *multiplicative inverse* of a modulo m (often denoted $a^{-1} \pmod{m}$).

§ Corollary 2.8 (When inverses exist) The congruence $ax \equiv 1 \pmod{m}$ has a solution iff $\gcd(a, m) = 1$. In that case the inverse is unique modulo m .

Proof. Apply Theorem ?? with $b = 1$. If $d = \gcd(a, m) > 1$ then $d \nmid 1$ so there is no solution. If $d = 1$ then there is exactly one solution modulo m . $\square$

§ Example (solve $5x \equiv 12 \pmod{16}$) Since $\gcd(5, 16) = 1$, the congruence has a unique solution modulo 16 by Corollary ??. One checks that $5^{-1} \pmod{16} = 13$ because $5 \cdot 13 = 65 \equiv 1 \pmod{16}$. Multiplying by 13 gives

$$x \equiv 13 \cdot 12 \pmod{16}.$$

Compute $13 \cdot 12 = 156 \equiv 12 \pmod{16}$ (since $156 - 144 = 12$), so $x \equiv 12 \pmod{16}$.

0.7 L7: 2.3 Chinese Remainder Thm, 2.4 Wilson's Thm, 2.5 Fermat's Little Thm

L7: The Chinese Remainder Theorem, Wilson's Theorem, Fermat's Little Theorem

Example (motivation). Find a positive integer x with

$$x \equiv 2 \pmod{3}, \quad x \equiv 1 \pmod{4}, \quad x \equiv 3 \pmod{5}.$$

This is a system of linear congruences in one variable.

§ Theorem 2.9 (Chinese Remainder Theorem) Let $m_1, m_2, \dots, m_n$ be pairwise relatively prime positive integers, and let $b_1, \dots, b_n$ be arbitrary integers. Then the system

$$x \equiv b_i \pmod{m_i} \quad (i = 1, \dots, n)$$

has a solution, and any two solutions are congruent modulo $M := m_1 m_2 \cdots m_n$. Hence the solution is unique modulo M .

Proof. Let $M := \prod_{i=1}^n m_i$ and define $M_i := M/m_i$. Since $(M_i, m_i) = 1$ by the pairwise coprimality hypothesis, by ?? (existence of inverses when gcd is 1) there exists x_i with

$$M_i x_i \equiv 1 \pmod{m_i}.$$

Put

$$x := \sum_{i=1}^n b_i M_i x_i.$$

For a fixed index i , reduce modulo m_i : every term $b_j M_j x_j$ with $j \neq i$ is a multiple of m_i (since $m_i \mid M_j$ for $j \neq i$), while $M_i x_i \equiv 1 \pmod{m_i}$ by construction. Thus $x \equiv b_i \pmod{m_i}$ for each i , so x is a solution.

If x' is another solution, then $m_i \mid (x - x')$ for every i , and since the m_i are pairwise coprime it follows that $M \mid (x - x')$. Hence $x \equiv x' \pmod{M}$, proving uniqueness modulo M . $\square$

Example (continued). Here $m_1 = 3, m_2 = 4, m_3 = 5$, so $M = 60$ and

$$M_1 = 60/3 = 20, \quad M_2 = 60/4 = 15, \quad M_3 = 60/5 = 12.$$

Solve $M_i x_i \equiv 1 \pmod{m_i}$:

$$20x_1 \equiv 1 \pmod{3} \implies 2x_1 \equiv 1 \pmod{3} \implies x_1 \equiv 2 \pmod{3},$$

so choose $x_1 = 2$.

$$15x_2 \equiv 1 \pmod{4} \implies 3x_2 \equiv 1 \pmod{4} \implies x_2 \equiv 3 \pmod{4},$$

so choose $x_2 = 3$.

$$12x_3 \equiv 1 \pmod{5} \implies 2x_3 \equiv 1 \pmod{5} \implies x_3 \equiv 3 \pmod{5},$$

so choose $x_3 = 3$.

Now

$$x = b_1 M_1 x_1 + b_2 M_2 x_2 + b_3 M_3 x_3 = 2 \cdot 20 \cdot 2 + 1 \cdot 15 \cdot 3 + 3 \cdot 12 \cdot 3 = 80 + 45 + 108 = 233.$$

Hence $x \equiv 233 \equiv 53 \pmod{60}$, and the least positive solution is 53.

§ Lemma 2.10 (Self-inverse modulo a prime) Let p be a prime and $a \in \mathbb{Z}$. Then a is its own inverse modulo p (i.e. $a^2 \equiv 1 \pmod{p}$) if and only if $a \equiv \pm 1 \pmod{p}$.

Proof. If $a^2 \equiv 1 \pmod{p}$ then $p \mid (a-1)(a+1)$. Since p is prime, $p \mid (a-1)$ or $p \mid (a+1)$, so $a \equiv 1$ or $a \equiv -1 \pmod{p}$. The converse is immediate. $\square$

§ Theorem 2.11 (Wilson's Theorem) Let p be a prime. Then

$$(p-1)! \equiv -1 \pmod{p}.$$

Proof. The result is trivial for $p = 2, 3$, so assume $p \geq 5$. The integers $1, 2, \dots, p-1$ can be paired into multiplicative inverses modulo p . By Lemma ??, the only elements that are their own inverses are 1 and $p-1$. Thus, aside from 1 and $p-1$, the remaining $p-3$ elements pair off, and each pair multiplies to 1 modulo p . Therefore

$$(p-1)! \equiv 1 \cdot (\text{product of inverse-pairs}) \cdot (p-1) \equiv 1 \cdot 1 \cdots 1 \cdot (-1) \equiv -1 \pmod{p},$$

as required. $\square$

§ Remark (converse) The converse is also true: if $n > 1$ and $(n-1)! \equiv -1 \pmod{n}$ then n must be prime.

§ Proposition 2.12 (Converse to Wilson's Theorem) Let $n > 1$ be an integer. If $(n-1)! \equiv -1 \pmod{n}$ then n is prime.

Proof. Suppose n is composite, so $n = ab$ with $1 < a \leq b < n$. Then $a \mid (n-1)!$ because $(n-1)!$ contains the factor a , and hence $(n-1)! \equiv 0 \pmod{a}$. The hypothesis $(n-1)! \equiv -1 \pmod{n}$ implies $(n-1)! \equiv -1 \pmod{a}$ (since $a \mid n$), so $-1 \equiv 0 \pmod{a}$, which is impossible. Therefore no such nontrivial factor a exists and n must be prime. $\square$

§ Remark (Wilson primes) A prime p is called a *Wilson prime* if $(p-1)! \equiv -1 \pmod{p^2}$. Only three Wilson primes are known: 5, 13, 563. It is unknown whether infinitely many (or only finitely many) Wilson primes exist.

§ Theorem 2.13 (Fermat's Little Theorem) Let p be a prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof. Consider the $p-1$ integers $a, 2a, \dots, (p-1)a$. None of these is congruent to 0 $\pmod{p}$ (since $p \nmid a$), and no two are congruent modulo p (if $ia \equiv ja \pmod{p}$ with $1 \leq i < j \leq p-1$, multiply by the inverse of a modulo p , which exists by ??, to deduce $i \equiv j \pmod{p}$, impossible). Hence the residues of $a, 2a, \dots, (p-1)a$ modulo p are a permutation of $1, 2, \dots, p-1$. Multiplying all these congruences yields

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

Since $p \nmid (p-1)!$ (true for prime p), we may cancel $(p-1)!$ from both sides to obtain $a^{p-1} \equiv 1 \pmod{p}$. (Note: this cancellation step uses only that $\gcd((p-1)!, p) = 1$; Wilson's Theorem is not required.) $\square$

0.8 L8: 2.5 Pseudoprimes, 2.6 Euler's Thm, 3.1 Arithmetic functions/multiplicativity

L8: Corollaries to Fermat's Little Theorem, Pseudoprimes, Euler's Theorem, Arithmetic Functions

§ Corollary 2.14 (Inverse modulo a prime) Let p be a prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Then a^{p-2} is the multiplicative inverse of a modulo p ; that is,

$$a \cdot a^{p-2} \equiv 1 \pmod{p}.$$

Proof. By Fermat's Little Theorem (Theorem ??), $a^{p-1} \equiv 1 \pmod{p}$. Multiply both sides by a^{-1} if you like, or simply rewrite $a^{p-1} = a \cdot a^{p-2}$ to get $a \cdot a^{p-2} \equiv 1 \pmod{p}$. Thus a^{p-2} is an inverse of a modulo p . $\square$

§ Remark (application of Corollary ??) Corollary ?? is useful for solving linear congruences $ax \equiv b \pmod{p}$ when p is prime and $p \nmid a$: multiply both sides by a^{p-2} to obtain the solution.

§ Corollary 2.15 (Fresh form of Fermat) Let p be prime and $a \in \mathbb{Z}$. Then

$$a^p \equiv a \pmod{p}.$$

Proof. If $p \mid a$ the congruence is trivial. If $p \nmid a$, multiply Fermat's Little Theorem $a^{p-1} \equiv 1 \pmod{p}$ by a to get $a^p \equiv a \pmod{p}$. $\square$

§ Corollary 2.16 (case $a = 2$) For every prime p ,

$$2^p \equiv 2 \pmod{p}.$$

Proof. Take $a = 2$ in Corollary ??.

§ Question (converse for base 2) Is the converse of Corollary ?? true? That is, if $n > 1$ and $2^n \equiv 2 \pmod{n}$, must n be prime?

§ Answer and counterexample (341) No. Example: $n = 341 = 11 \cdot 31$ is composite but satisfies $2^{341} \equiv 2 \pmod{341}$. Since $(11, 31) = 1$, it suffices to check modulo 11 and 31.

By Fermat's Little Theorem,

$$2^{10} \equiv 1 \pmod{11} \Rightarrow 2^{341} \equiv (2^{10})^{34} \cdot 2 \equiv 1^{34} \cdot 2 \equiv 2 \pmod{11},$$

and

$$2^{30} \equiv 1 \pmod{31} \Rightarrow 2^{341} \equiv (2^{30})^{11} \cdot 2^{11} \equiv 1^{11} \cdot 2^{11} \pmod{31}.$$

Since $2^5 \equiv -1 \pmod{31}$ (or compute directly), we have $2^{11} \equiv 2 \pmod{31}$, so $2^{341} \equiv 2 \pmod{31}$. Hence $2^{341} \equiv 2 \pmod{341}$ although 341 is composite.

§ Definition (pseudoprime to base 2) A composite integer $n > 1$ is called a (*base-2*) *pseudoprime* if

$$2^n \equiv 2 \pmod{n}.$$

§ Remark (pseudoprimes) 341 is the smallest base-2 pseudoprime. There are infinitely many pseudoprimes and they arise for different bases; pseudoprimes show that the naive converse of Fermat's Little Theorem is false.

§ Definition (Euler’s phi-function) For $n \in \mathbb{Z}_{>0}$ the Euler totient function $\phi(n)$ is the number of integers $1 \leq x \leq n$ with $\gcd(x, n) = 1$; equivalently,

$$\phi(n) := \#\{x \in \{1, 2, \dots, n\} : \gcd(x, n) = 1\}.$$

§ Examples (values of ϕ)

$$\phi(12) = 4 \quad (\text{the classes } 1, 5, 7, 11), \quad \phi(14) = 6 \quad (\text{the classes } 1, 3, 5, 9, 11, 13),$$

and for a prime p , $\phi(p) = p - 1$ (classes $1, 2, \dots, p - 1$).

§ Theorem 2.17 (Euler’s Theorem) Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. Then

$$a^{\phi(m)} \equiv 1 \pmod{m}.$$

Proof. Let $r_1, r_2, \dots, r_{\phi(m)}$ be the positive integers $\leq m$ that are relatively prime to m . Since $\gcd(a, m) = 1$, multiplication by a permutes this reduced residue system: the numbers $ar_1, \dots, ar_{\phi(m)}$ are all distinct modulo m and each is relatively prime to m (see Lemma 1.5 and Lemma 1.14 referenced earlier). Hence the set $\{ar_i \bmod m\}$ is the same set of residues as $\{r_i\}$. Multiplying all residues yields

$$a^{\phi(m)} \prod_{i=1}^{\phi(m)} r_i \equiv \prod_{i=1}^{\phi(m)} r_i \pmod{m}.$$

Since $\gcd(\prod_i r_i, m) = 1$, we may cancel the product to obtain $a^{\phi(m)} \equiv 1 \pmod{m}$. □

§ Remark (special case) Fermat’s Little Theorem is the special case $m = p$ prime, since $\phi(p) = p - 1$.

§ Definition (reduced residue system) A set of $\phi(m)$ integers each relatively prime to m and pairwise incongruent modulo m is called a *reduced residue system* modulo m .

§ Example (reduced residue systems) $\{1, 5, 7, 11\}$ is a reduced residue system modulo 12. If a is coprime to m , then $\{ar_1, \dots, ar_{\phi(m)}\}$ is also a reduced residue system modulo m (this was the idea used in the proof of Euler’s Theorem).

§ Corollary 2.19 (inverse via Euler) Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. Then $a^{\phi(m)-1}$ is the multiplicative inverse of a modulo m .

Proof. From $a^{\phi(m)} \equiv 1 \pmod{m}$ multiply by a^{-1} (or rewrite as $a \cdot a^{\phi(m)-1} \equiv 1$) to see $a^{\phi(m)-1}$ is an inverse of a modulo m . □

§ Remark (use of Corollary ??) Corollary ?? helps solve linear congruences $ax \equiv b \pmod{m}$ when $\gcd(a, m) = 1$: multiply both sides by $a^{\phi(m)-1}$ to obtain the solution.

§ Definition (arithmetic function) An *arithmetic function* is a function $f : \mathbb{Z}_{>0} \rightarrow \mathbb{C}$ (or $\mathbb{R}$, or $\mathbb{Z}$). Examples: the Euler ϕ -function, the divisor-counting function $d(n)$, the sum-of-divisors function $\sigma(n)$, etc.

§ Definition (multiplicative vs completely multiplicative) An arithmetic function f is *multiplicative* if $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$. It is *completely multiplicative* if $f(mn) = f(m)f(n)$ for all positive integers m, n (no coprimality required).

§ Proposition (values determined by prime powers) If f is multiplicative and

$$n = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}$$

is the prime factorization of n , then

$$f(n) = \prod_{i=1}^r f(p_i^{a_i}).$$

If f is completely multiplicative, then

$$f(n) = \prod_{i=1}^r f(p_i)^{a_i}.$$

Proof. Write n as a product of relatively prime prime-power factors and apply multiplicativity repeatedly. □

§ Remark (implication) Thus knowing a multiplicative arithmetic function on prime powers (or a completely multiplicative function on primes) determines it everywhere.

0.9 L9: 3.1 Arithmetic functions/multiplicativity, 3.2 Euler phi function

L9: Arithmetic Functions and Multiplicativity

§ Summation over divisors: notation By

$$\sum_{d|n} f(d)$$

we mean the sum of $f(d)$ over all *positive* divisors d of n . For example,

$$\sum_{d|12} f(d) = f(1) + f(2) + f(3) + f(4) + f(6) + f(12).$$

§ Multiplicative functions and divisor sums Let f be an arithmetic function and define, for $n \in \mathbb{Z}_{>0}$,

$$F(n) := \sum_{d|n} f(d).$$

If f is multiplicative then F is multiplicative.

Proof. Let $m, n \in \mathbb{Z}_{>0}$ with $\gcd(m, n) = 1$. We must show $F(mn) = F(m)F(n)$.

Claim: Every positive divisor d of mn can be written uniquely as $d = d_1 d_2$ with $d_1 | m$, $d_2 | n$ and $\gcd(d_1, d_2) = 1$.

Proof of claim: Write the prime factorizations

$$m = \prod_{i=1}^r p_i^{a_i}, \quad n = \prod_{j=1}^s q_j^{b_j},$$

where the primes p_i and q_j are distinct because $\gcd(m, n) = 1$. Then any divisor d of mn has the form

$$d = \prod_{i=1}^r p_i^{e_i} \prod_{j=1}^s q_j^{f_j},$$

with $0 \leq e_i \leq a_i$ and $0 \leq f_j \leq b_j$. Set

$$d_1 = \prod_{i=1}^r p_i^{e_i}, \quad d_2 = \prod_{j=1}^s q_j^{f_j}.$$

Clearly $d_1 | m$, $d_2 | n$, $\gcd(d_1, d_2) = 1$, and $d = d_1 d_2$. Uniqueness follows from uniqueness of prime factorization. $\square$

Using the claim,

$$F(mn) = \sum_{d|mn} f(d) = \sum_{\substack{d_1|m \\ d_2|n}} f(d_1 d_2).$$

Since f is multiplicative and $\gcd(d_1, d_2) = 1$ for every pair occurring in the sum, we have $f(d_1 d_2) = f(d_1)f(d_2)$. Thus

$$F(mn) = \sum_{d_1|m} \sum_{d_2|n} f(d_1)f(d_2) = \left(\sum_{d_1|m} f(d_1) \right) \left(\sum_{d_2|n} f(d_2) \right) = F(m)F(n).$$

This proves that F is multiplicative. $\square$

§ Example: verification for $m = 3$, $n = 4$ Let $m = 3$ and $n = 4$. The positive divisors are

$$\{d_1 | 3\} = \{1, 3\}, \quad \{d_2 | 4\} = \{1, 2, 4\}.$$

All divisors of 12 are the products $d_1 d_2$ with $d_1 \in \{1, 3\}$ and $d_2 \in \{1, 2, 4\}$; hence

$$\sum_{d|12} f(d) = \sum_{d_1|3} \sum_{d_2|4} f(d_1 d_2) = \sum_{d_1|3} \sum_{d_2|4} f(d_1) f(d_2)$$

(using multiplicativity of f), so

$$\sum_{d|12} f(d) = \left(\sum_{d_1|3} f(d_1) \right) \left(\sum_{d_2|4} f(d_2) \right) = F(3)F(4).$$

§ The Euler φ -function is multiplicative The Euler totient function $\varphi(n)$ (the number of integers in $\{1, 2, \dots, n\}$ relatively prime to n) is multiplicative: if $\gcd(m, n) = 1$ then

$$\varphi(mn) = \varphi(m) \varphi(n).$$

Proof. List the integers $1, 2, \dots, mn$ in an $m \times n$ array with rows indexed by $i = 1, \dots, m$ and columns by $k = 0, \dots, n-1$ as

$$\begin{array}{cccccc} 1 & m+1 & 2m+1 & \cdots & (n-1)m+1 \\ 2 & m+2 & 2m+2 & \cdots & (n-1)m+2 \\ \vdots & \vdots & \vdots & & \vdots \\ m & m+m & 2m+m & \cdots & (n-1)m+m \end{array}$$

Consider the i -th row

$$i, m+i, 2m+i, \dots, (n-1)m+i.$$

If $\gcd(i, m) > 1$ then every entry of that row shares a common factor with m , and hence cannot be relatively prime to mn . Thus only rows with $\gcd(i, m) = 1$ can contain integers relatively prime to mn . There are precisely $\varphi(m)$ such rows.

Fix a row with $\gcd(i, m) = 1$. The n entries of that row are distinct modulo n : if $jm+i \equiv km+i \pmod{n}$ with $0 \leq j, k \leq n-1$, then $(j-k)m \equiv 0 \pmod{n}$. Since $\gcd(m, n) = 1$, m has a multiplicative inverse modulo n , so $j \equiv k \pmod{n}$, hence $j = k$. Thus the row is a complete residue system modulo n , so exactly $\varphi(n)$ of its entries are relatively prime to n . Because those entries are relatively prime to both m and n , they are relatively prime to mn .

Therefore each of the $\varphi(m)$ admissible rows contains exactly $\varphi(n)$ integers relatively prime to mn . Thus

$$\varphi(mn) = \varphi(m) \varphi(n).$$

□

§ Values of φ at prime powers Let p be prime and $a \in \mathbb{Z}_{>0}$. Then

$$\varphi(p^a) = p^a - p^{a-1} = p^a \left(1 - \frac{1}{p} \right).$$

Proof. Among the p^a integers $1, 2, \dots, p^a$, those not coprime to p^a are exactly the multiples of p :

$$p, 2p, 3p, \dots, p^{a-1}p,$$

and there are p^{a-1} such multiples. Hence

$$\varphi(p^a) = p^a - p^{a-1} = p^a \left(1 - \frac{1}{p} \right).$$

□

§ **Product formula for $\varphi(n)$** For $n \in \mathbb{Z}_{>0}$,

$$\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right),$$

where the product is taken over the distinct prime divisors p of n .

Proof. If $n = 1$ the result holds (empty product = 1). For $n > 1$, write the prime factorization

$$n = \prod_{i=1}^r p_i^{a_i}.$$

By Theorems ?? and ??,

$$\varphi(n) = \prod_{i=1}^r \varphi(p_i^{a_i}) = \prod_{i=1}^r p_i^{a_i} \left(1 - \frac{1}{p_i}\right) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

□

§ **Example:** $\varphi(504)$ Since $504 = 2^3 \cdot 3^2 \cdot 7$, we get

$$\varphi(504) = 504 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{7}\right).$$

Compute:

$$\left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{7}\right) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{6}{7} = \frac{2}{7},$$

so $\varphi(504) = 504 \cdot \frac{2}{7} = 144$.

§ **Gauss's identity for φ** For $n \in \mathbb{Z}_{>0}$,

$$\sum_{d|n} \varphi(d) = n.$$

Proof. For each positive divisor d of n define

$$S_d := \{m \in \mathbb{Z} : 1 \leq m \leq n, \gcd(m, n) = d\}.$$

Note that $\gcd(m, n) = d$ if and only if $\gcd(m/d, n/d) = 1$, so the number of integers in S_d equals $\varphi(n/d)$. The sets S_d (as d runs over the positive divisors of n) partition the set $\{1, 2, \dots, n\}$; hence

$$n = \sum_{d|n} |S_d| = \sum_{d|n} \varphi(n/d).$$

Reindexing the sum by replacing d with n/d shows

$$n = \sum_{d|n} \varphi(d),$$

as claimed. □

§ **Example:** $n = 12$ For $n = 12$ the divisors and corresponding sets are

$$\begin{aligned}
S_1 &= \{1, 5, 7, 11\}, & |S_1| &= \varphi(12) = 4, \\
S_2 &= \{2, 10\}, & |S_2| &= \varphi(6) = 2, \\
S_3 &= \{3, 9\}, & |S_3| &= \varphi(4) = 2, \\
S_4 &= \{4, 8\}, & |S_4| &= \varphi(3) = 2, \\
S_6 &= \{6\}, & |S_6| &= \varphi(2) = 1, \\
S_{12} &= \{12\}, & |S_{12}| &= \varphi(1) = 1.
\end{aligned}$$

Summing gives

$$\sum_{d|12} \varphi(d) = 4 + 2 + 2 + 2 + 1 + 1 = 12,$$

confirming Gauss's identity.

0.10 L10: 3.3 Divisor function, 3.4 Sum of divisors, 3.5 Perfect numbers, 3.6 Möbius function

L10: The number of positive divisors function

§ The number of positive divisors function Let $n \in \mathbb{Z}_{>0}$. The number of positive divisors of n (the divisor-count function) is the function

$$V(n) := \#\{d \in \mathbb{Z}_{>0} : d \mid n\}.$$

§ Multiplicativity of the divisor-count function The function $V(n)$ is multiplicative: if $\gcd(m, n) = 1$ then

$$V(mn) = V(m)V(n).$$

Proof. Write

$$V(n) = \sum_{d \mid n} 1,$$

the sum of the constant arithmetic function $f(d) \equiv 1$ over positive divisors of n . The constant function $f(d) = 1$ is multiplicative, so by Theorem ?? (divisor-sum preserving multiplicativity) the divisor-sum $V(n)$ is multiplicative. $\square$

§ Values of V at prime powers Let p be prime and $a \in \mathbb{Z}_{\geq 0}$. Then

$$V(p^a) = a + 1.$$

Proof. The positive divisors of p^a are exactly

$$1, p, p^2, \dots, p^a,$$

so there are $a + 1$ of them. $\square$

§ Prime-power formula for $V(n)$ If

$$n = \prod_{i=1}^r p_i^{a_i}$$

is the prime factorization of n (distinct primes p_i , exponents $a_i \geq 0$), then

$$V(n) = \prod_{i=1}^r (a_i + 1).$$

Proof. Since V is multiplicative (Theorem ??), and each $V(p_i^{a_i}) = a_i + 1$ (Theorem ??), multiplicativity yields

$$V(n) = \prod_{i=1}^r V(p_i^{a_i}) = \prod_{i=1}^r (a_i + 1).$$

$\square$

§ Example: $V(504)$ If $504 = 2^3 \cdot 3^2 \cdot 7^1$ then

$$V(504) = (3 + 1)(2 + 1)(1 + 1) = 4 \cdot 3 \cdot 2 = 24.$$

§ The sum-of-divisors function σ For $n \in \mathbb{Z}_{>0}$ define

$$\sigma(n) := \sum_{d|n} d,$$

the sum of the positive divisors of n .

§ Multiplicativity of σ The function $\sigma(n)$ is multiplicative: if $\gcd(m, n) = 1$ then

$$\sigma(mn) = \sigma(m)\sigma(n).$$

Proof. This follows from Theorem ?? by observing that the arithmetic function $f(d) = d$ is multiplicative: if $\gcd(a, b) = 1$ then $(ab) = a \cdot b$ and summing multiplicatively gives the result. (Equivalently, check $f(ab) = ab = f(a)f(b)$ for coprime a, b .) $\square$

§ Values of σ at prime powers Let p be prime and $a \in \mathbb{Z}_{\geq 0}$. Then

$$\sigma(p^a) = 1 + p + \cdots + p^a = \frac{p^{a+1} - 1}{p - 1}.$$

Proof. The divisors of p^a are $1, p, \dots, p^a$, so the sum is the finite geometric series $1 + p + \cdots + p^a = (p^{a+1} - 1)/(p - 1)$. $\square$

§ Prime-power product formula for $\sigma(n)$ If $n = \prod_{i=1}^r p_i^{a_i}$ then

$$\sigma(n) = \prod_{i=1}^r \frac{p_i^{a_i+1} - 1}{p_i - 1}.$$

Proof. By multiplicativity (Theorem ??) and Theorem ?? applied to each prime power factor. $\square$

§ Example: $\sigma(504)$ For $504 = 2^3 \cdot 3^2 \cdot 7$,

$$\sigma(504) = \left(\frac{2^4 - 1}{2 - 1}\right) \left(\frac{3^3 - 1}{3 - 1}\right) \left(\frac{7^2 - 1}{7 - 1}\right) = 15 \cdot 13 \cdot 8 = 1560.$$

§ Perfect numbers (definition) A positive integer n is *perfect* if $\sigma(n) = 2n$. Equivalently, the sum of the proper divisors of n (i.e., divisors $< n$) equals n .

§ Classical characterization of even perfect numbers An integer $n > 0$ is an even perfect number if and only if

$$n = 2^{p-1}(2^p - 1),$$

where p is prime and $2^p - 1$ is prime (a Mersenne prime).

Proof. We give both directions (Euler $\Rightarrow$ and Euclid $\Leftarrow$).

($\Leftarrow$) (**Euclid**). Suppose p and $2^p - 1$ are prime and set $n = 2^{p-1}(2^p - 1)$. Using multiplicativity of σ ,

$$\sigma(n) = \sigma(2^{p-1})\sigma(2^p - 1) = \frac{2^p - 1}{2 - 1} \cdot (1 + 2^p - 1) = (2^p - 1) \cdot 2^p = 2n,$$

since $\sigma(2^{p-1}) = 1 + 2 + \cdots + 2^{p-1} = (2^p - 1)$ and $\sigma(2^p - 1) = 1 + (2^p - 1) = 2^p$ (because $2^p - 1$ is prime). Hence n is perfect.

($\Rightarrow$) (**Euler**). Let n be an even perfect number. Write $n = 2^r m$ with $r \geq 1$ and m odd. Then multiplicativity yields

$$\sigma(n) = \sigma(2^r)\sigma(m) = (2^{r+1} - 1)\sigma(m).$$

Since n is perfect, $\sigma(n) = 2n = 2^{r+1}m$. Thus

$$(2^{r+1} - 1)\sigma(m) = 2^{r+1}m.$$

Because $\gcd(2^{r+1} - 1, 2^{r+1}) = 1$, it follows that $2^{r+1} \mid \sigma(m)$. Hence write $\sigma(m) = 2^{r+1}c$ for some $c \in \mathbb{Z}_{>0}$, and substituting back gives

$$(2^{r+1} - 1)2^{r+1}c = 2^{r+1}m \implies m = (2^{r+1} - 1)c.$$

We now show $c = 1$. If $c > 1$ then m has at least the divisors $1, c, m$ with $1 < c < m$, so

$$\sigma(m) \geq 1 + c + m = 1 + c + (2^{r+1} - 1)c = 1 + (2^{r+1})c,$$

whence $\sigma(m) \geq 1 + 2^{r+1}c$, contradicting $\sigma(m) = 2^{r+1}c$. Thus $c = 1$ and $m = 2^{r+1} - 1$. Then $\sigma(m) = m + 1 = 2^{r+1}$, so m is prime and $m = 2^{r+1} - 1$ is a Mersenne prime. Setting $p = r + 1$ gives $n = 2^{p-1}(2^p - 1)$ with p prime and $2^p - 1$ prime, as required. $\square$

§ Remarks on perfect numbers - This theorem establishes a bijection between Mersenne primes and even perfect numbers. It is unknown whether there are infinitely many Mersenne primes (hence unknown whether there are infinitely many even perfect numbers). - It is also unknown whether any odd perfect numbers exist; the conjecture "every perfect number is even" remains open.

§ The Möbius function μ The Möbius function $\mu : \mathbb{Z}_{>0} \rightarrow \{-1, 0, 1\}$ is defined by

$$\mu(n) = \begin{cases} 1, & n = 1, \\ 0, & \text{if } p^2 \mid n \text{ for some prime } p, \\ (-1)^r, & \text{if } n = p_1 p_2 \cdots p_r \text{ is a product of } r \text{ distinct primes.} \end{cases}$$

§ Examples for μ Since $504 = 2^3 \cdot 3^2 \cdot 7$ contains a squared prime factor, $\mu(504) = 0$. Since $30 = 2 \cdot 3 \cdot 5$ is a product of three distinct primes, $\mu(30) = (-1)^3 = -1$.

§ Multiplicativity of μ The Möbius function $\mu(n)$ is multiplicative: if $\gcd(m, n) = 1$ then $\mu(mn) = \mu(m)\mu(n)$.

Proof. If either m or n is divisible by a square of a prime then so is mn , and both sides are 0. Otherwise write $m = \prod_{i=1}^r p_i$ and $n = \prod_{j=1}^s q_j$ as products of distinct primes (with the two lists disjoint by coprimality). Then

$$\mu(mn) = (-1)^{r+s} = (-1)^r (-1)^s = \mu(m)\mu(n).$$

$\square$

§ Möbius summatory identity (Proposition) For $n \in \mathbb{Z}_{>0}$,

$$\sum_{d \mid n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

Proof. Define $F(n) = \sum_{d \mid n} \mu(d)$. The function F is multiplicative since it is a divisor-sum of a multiplicative function μ (Theorems ?? and ??). Thus F is determined by its values on prime powers. Clearly $F(1) = 1$. If p is prime and $a \geq 1$ then

$$F(p^a) = \mu(1) + \mu(p) + \mu(p^2) + \cdots + \mu(p^a) = 1 + (-1) + 0 + \cdots + 0 = 0.$$

Hence $F(n) = 0$ for every $n > 1$ (by multiplicativity and prime-power vanishing), proving the claim. $\square$

§ Example: $n = 12$ (check) For $n = 12$ the divisors are 1, 2, 3, 4, 6, 12 and

$$\mu(1) + \mu(2) + \mu(3) + \mu(4) + \mu(6) + \mu(12) = 1 + (-1) + (-1) + 0 + 1 + 0 = 0,$$

consistent with Proposition ??.

0.11 L11: 3.6 Möbius inversion, 4.1 Quadratic residues

§ Möbius Inversion Let f and g be arithmetic functions (functions $\mathbb{Z}_{>0} \rightarrow \mathbb{C}$). Then

$$f(n) = \sum_{d|n} g(d) \quad \text{for all } n \geq 1$$

if and only if

$$g(n) = \sum_{d|n} \mu(d) f\left(\frac{n}{d}\right) = \sum_{d|n} \mu\left(\frac{n}{d}\right) f(d) \quad \text{for all } n \geq 1,$$

where μ denotes the Möbius function.

Proof. We prove both directions.

(“ $\Rightarrow$ ”) Assume $f(n) = \sum_{d|n} g(d)$ for every $n \geq 1$. Then for each fixed n ,

$$\sum_{d|n} \mu(d) f\left(\frac{n}{d}\right) = \sum_{d|n} \mu(d) \sum_{c|\frac{n}{d}} g(c).$$

The double sum can be rearranged by letting c run over divisors of n and collecting terms in d with $d | n/c$:

$$\sum_{c|n} g(c) \sum_{d|n/c} \mu(d).$$

By the well-known identity (see Proposition ??) the inner sum $\sum_{d|m} \mu(d)$ equals 1 if $m = 1$ and 0 otherwise. Hence the inner sum is 1 exactly when $n/c = 1$, i.e. $c = n$, and is 0 for all other c . Thus the whole expression reduces to $g(n)$, proving the claimed formula for $g(n)$.

(“ $\Leftarrow$ ”) Conversely, assume $g(n) = \sum_{d|n} \mu(n/d) f(d)$ for all n . Then

$$\sum_{d|n} g(d) = \sum_{d|n} \sum_{c|d} \mu(d/c) f(c).$$

Rearrange by summing first over $c | n$ and then over d with $c | d | n$:

$$\sum_{c|n} f(c) \sum_{\substack{d \\ c|d, d|n}} \mu(d/c) = \sum_{c|n} f(c) \sum_{m|(n/c)} \mu(m).$$

Again, by Proposition ?? the inner sum is 1 if $n/c = 1$ (i.e. $c = n$) and 0 otherwise. Therefore the sum reduces to $f(n)$, proving that $f(n) = \sum_{d|n} g(d)$ for every n . $\square$

§ Remark The Möbius inversion formula is exactly the statement that μ is the convolutional inverse of the constant-1 function with respect to Dirichlet convolution. Equivalently, if $f = g * 1$ (Dirichlet convolution), then $g = f * \mu$. (See Homework 4 for exercises on this point.)

Example. Let $f(n) = n$ for all $n \geq 1$. By the well-known identity

$$n = \sum_{d|n} \varphi(d),$$

we may take $g(d) = \varphi(d)$. Applying Möbius inversion yields

$$\varphi(n) = \sum_{d|n} \mu(d) \frac{n}{d} = \sum_{d|n} \mu\left(\frac{n}{d}\right) d.$$

Example. Let $\tau(n)$ denote the number of positive divisors of n (in your notes this was written as $v(n)$; I replaced it by the standard notation $\tau(n)$ and noted this change above). Then

$$\tau(n) = \sum_{d|n} 1,$$

so $g(d) = 1$. By Möbius inversion we obtain the identity

$$1 = \sum_{d|n} \mu(d) \tau\left(\frac{n}{d}\right) = \sum_{d|n} \mu\left(\frac{n}{d}\right) \tau(d).$$

Example. Let $\sigma(n) = \sum_{d|n} d$ be the sum-of-divisors function. Then $g(d) = d$ and Möbius inversion gives

$$n = \sum_{d|n} \mu(d) \sigma\left(\frac{n}{d}\right) = \sum_{d|n} \mu\left(\frac{n}{d}\right) \sigma(d).$$

Quadratic residues

In the previous sections we studied linear congruences of the form $ax \equiv b \pmod{m}$. We now turn our attention to *quadratic* congruences; classically one studies congruences of the form

$$x^2 \equiv a \pmod{p},$$

especially in the case where p is an odd prime. A central tool is the Legendre symbol $\left(\frac{a}{p}\right)$ (introduced later), which encodes whether a is a quadratic residue modulo p .

§ Quadratic residue modulo m Let $a, m \in \mathbb{Z}$ with $m > 0$ and $\gcd(a, m) = 1$. We say that a is a *quadratic residue modulo m* if the congruence

$$x^2 \equiv a \pmod{m}$$

has a solution $x \in \mathbb{Z}$. Otherwise a is a *quadratic non-residue modulo m* .

Example (quadratic residues modulo 11). Compute $x^2 \pmod{11}$ for $x = 1, 2, \dots, 10$:

$$\begin{aligned} 1^2 &\equiv 1, & 2^2 &\equiv 4, & 3^2 &\equiv 9, \\ 4^2 &\equiv 5, & 5^2 &\equiv 3, & 6^2 &\equiv 3, \\ 7^2 &\equiv 5, & 8^2 &\equiv 9, & 9^2 &\equiv 4, \\ 10^2 &\equiv 1 \end{aligned} \pmod{11}$$

Thus the distinct quadratic residues modulo 11 (among the reduced residue classes) are

$$1, 3, 4, 5, 9,$$

and the quadratic non-residues (modulo 11) are

$$2, 6, 7, 8, 10.$$

§ Proposition Let p be an odd prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Then the congruence

$$x^2 \equiv a \pmod{p}$$

has either no solution or exactly two incongruent solutions modulo p .

Proof. Suppose the congruence has a solution x_0 . Then $(-x_0)^2 \equiv x_0^2 \equiv a$, so $-x_0$ is also a solution. We claim $x_0 \not\equiv -x_0 \pmod{p}$. Indeed, if $x_0 \equiv -x_0 \pmod{p}$ then $2x_0 \equiv 0 \pmod{p}$. Since p is odd, $\gcd(2, p) = 1$, so this implies $x_0 \equiv 0 \pmod{p}$, and hence $a \equiv x_0^2 \equiv 0 \pmod{p}$, contradicting the assumption $p \nmid a$. Thus x_0 and $-x_0$ are distinct modulo p .

Now suppose x_0 and x_1 are any two solutions. Then $x_0^2 \equiv x_1^2 \pmod{p}$, so $p \mid (x_0^2 - x_1^2) = (x_0 - x_1)(x_0 + x_1)$. Since p is prime, we must have $p \mid (x_0 - x_1)$ or $p \mid (x_0 + x_1)$. Therefore $x_0 \equiv x_1 \pmod{p}$ or $x_0 \equiv -x_1 \pmod{p}$. This shows there are at most two incongruent solutions. Combined with the earlier observation that if one solution exists then a second distinct one does too, we conclude there are exactly two incongruent solutions in the solvable case. $\square$

§ Corollary Let p be an odd prime and let $a \in \mathbb{Z}$ with $p \nmid a$. If $x^2 \equiv a \pmod{p}$ is solvable and x_0 is one solution, then the two incongruent solutions modulo p are x_0 and $-x_0 \equiv p - x_0$.

§ Proposition Let p be an odd prime. There are exactly $\frac{p-1}{2}$ distinct (incongruent) quadratic residues modulo p among the nonzero residue classes, and exactly $\frac{p-1}{2}$ quadratic non-residues among the nonzero classes.

Proof. Consider the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ of size $p-1$. For each $x \in (\mathbb{Z}/p\mathbb{Z})^\times$, the square x^2 is a nonzero quadratic residue. Note that $x^2 = (-x)^2$, so each nonzero quadratic residue arises from exactly two distinct elements of $(\mathbb{Z}/p\mathbb{Z})^\times$, namely x and $-x$. Therefore the set of distinct nonzero squares has size $(p-1)/2$. Since the set of nonzero residue classes has size $p-1$, the remaining $(p-1)/2$ classes are quadratic non-residues. This completes the proof. $\square$

0.12 L12: 4.2 Legendre symbol

L12: The Legendre Symbol

§ Definition 4.1 (Legendre symbol) Let p be an odd prime and let $a \in \mathbb{Z}$. The *Legendre symbol* $\left(\frac{a}{p}\right)$ is defined by

$$\left(\frac{a}{p}\right) := \begin{cases} 0, & \text{if } p \mid a, \\ 1, & \text{if } p \nmid a \text{ and } a \text{ is a quadratic residue modulo } p, \\ -1, & \text{if } p \nmid a \text{ and } a \text{ is a quadratic non-residue modulo } p. \end{cases}$$

§ Examples Working modulo 11, the nonzero quadratic residues are the squares

$$1^2 \equiv 1, \quad 2^2 \equiv 4, \quad 3^2 \equiv 9, \quad 4^2 \equiv 5, \quad 5^2 \equiv 3 \pmod{11},$$

so the quadratic residues modulo 11 are

$$1, 3, 4, 5, 9.$$

Hence

$$\left(\frac{1}{11}\right) = \left(\frac{3}{11}\right) = \left(\frac{4}{11}\right) = \left(\frac{5}{11}\right) = \left(\frac{9}{11}\right) = 1,$$

and the remaining nonzero residues

$$2, 6, 7, 8, 10$$

are quadratic non-residues modulo 11, so their Legendre symbols equal -1 :

$$\left(\frac{2}{11}\right) = \left(\frac{6}{11}\right) = \left(\frac{7}{11}\right) = \left(\frac{8}{11}\right) = \left(\frac{10}{11}\right) = -1.$$

§ Discussion

- The Legendre symbol $\left(\frac{a}{p}\right)$ tells whether the congruence $x^2 \equiv a \pmod{p}$ is solvable (value 1), not solvable (value -1), or whether $p \mid a$ (value 0).
- It does not provide the actual solutions of the congruence; if the symbol equals 1, one still must find the square roots mod p by other methods.

§ Example 4.2 Find $\left(\frac{3}{7}\right)$. By direct computation of squares modulo 7,

$$1^2 \equiv 1, \quad 2^2 \equiv 4, \quad 3^2 \equiv 2 \pmod{7},$$

so the nonzero quadratic residues modulo 7 are $\{1, 2, 4\}$ (there are $(7-1)/2 = 3$ of them). Therefore 3 is not a quadratic residue modulo 7, and

$$\left(\frac{3}{7}\right) = -1.$$

§ Theorem 4.4 (Euler's Criterion) Let p be an odd prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Then

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p},$$

and, since both sides take values in $\{\pm 1\}$, we may (and usually do) interpret this congruence as the equality

$$\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}} \pmod{p} \in \{1, -1\}.$$

Proof. First suppose $\left(\frac{a}{p}\right) = 1$. Then there exists x_0 with $x_0^2 \equiv a \pmod{p}$, so

$$a^{\frac{p-1}{2}} \equiv (x_0^2)^{\frac{p-1}{2}} \equiv x_0^{p-1} \equiv 1 \pmod{p}$$

by Fermat's little theorem. Hence $a^{(p-1)/2} \equiv 1 \pmod{p}$, which matches $\left(\frac{a}{p}\right) = 1$.

Now suppose $\left(\frac{a}{p}\right) = -1$. Consider the set $\{1, 2, \dots, p-1\}$. For each $i \in \{1, \dots, p-1\}$ there is a unique $j \in \{1, \dots, p-1\}$ with $ij \equiv a \pmod{p}$ (since multiplication by i is a permutation of the nonzero residues). If $i = j$ for some such pair then $i^2 \equiv a \pmod{p}$, contradicting that a is a non-residue; hence the elements $1, \dots, p-1$ can be paired into $(p-1)/2$ disjoint pairs whose products are congruent to a . Therefore

$$(p-1)! \equiv a^{\frac{p-1}{2}} \pmod{p}.$$

By Wilson's theorem $(p-1)! \equiv -1 \pmod{p}$, so $a^{(p-1)/2} \equiv -1 \pmod{p}$, which matches $\left(\frac{a}{p}\right) = -1$.

Combining both cases proves the statement. □

§ Example 4.3 (Using Euler's Criterion) Compute $\left(\frac{3}{7}\right)$ using Euler's criterion:

$$3^{\frac{7-1}{2}} = 3^3 = 27 \equiv 27 - 21 = 6 \equiv -1 \pmod{7},$$

so $\left(\frac{3}{7}\right) = -1$.

§ Proposition 4.5 Let p be an odd prime and let $a, b \in \mathbb{Z}$ with $p \nmid a$ and $p \nmid b$. Then

1. $\left(\frac{a^2}{p}\right) = 1$.
2. If $a \equiv b \pmod{p}$ then $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$.
3. $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right)$ (multiplicativity).

Proof. (a) If $p \nmid a$ then a is a solution of $x^2 \equiv a^2 \pmod{p}$, so $(a^2/p) = 1$.

(b) If $a \equiv b \pmod{p}$ then the congruences $x^2 \equiv a \pmod{p}$ and $x^2 \equiv b \pmod{p}$ are identical; hence solvability is the same for both, so the Legendre symbols agree.

(c) Using Euler's criterion,

$$\left(\frac{ab}{p}\right) \equiv (ab)^{\frac{p-1}{2}} \equiv a^{\frac{p-1}{2}} b^{\frac{p-1}{2}} \equiv \left(\frac{a}{p}\right) \left(\frac{b}{p}\right) \pmod{p}.$$

Both sides are in $\{\pm 1\}$, so the congruence implies equality:

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right).$$

□

§ **Example 4.4** Compute $\left(\frac{-11}{7}\right)$. We factor

$$\left(\frac{-11}{7}\right) = \left(\frac{-1}{7}\right) \left(\frac{11}{7}\right).$$

Since $11 \equiv 4 \pmod{7}$ and $4 = 2^2$ is a square, $(11/7) = (4/7) = 1$. Hence

$$\left(\frac{-11}{7}\right) = \left(\frac{-1}{7}\right).$$

By Theorem ?? below (or direct computation) $\left(\frac{-1}{7}\right) = -1$ because $7 \equiv 3 \pmod{4}$. Thus $\left(\frac{-11}{7}\right) = -1$.

§ **Further remarks on reduction to prime factors** Let p be an odd prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Write the integer a (up to sign) as a product of prime powers

$$a = \pm 2^{\alpha_0} q_1^{\alpha_1} q_2^{\alpha_2} \cdots q_r^{\alpha_r},$$

where the q_i are distinct odd primes different from p and the $\alpha_i \geq 0$ are integers. By multiplicativity (Proposition ??(c)) we have

$$\left(\frac{a}{p}\right) = \left(\frac{\pm 1}{p}\right) \left(\frac{2}{p}\right)^{\alpha_0} \left(\frac{q_1}{p}\right)^{\alpha_1} \cdots \left(\frac{q_r}{p}\right)^{\alpha_r}.$$

Hence it suffices to compute $\left(\frac{\pm 1}{p}\right)$, $\left(\frac{2}{p}\right)$, and $\left(\frac{q}{p}\right)$ for an odd prime $q \neq p$.

§ Plan / Roadmap

- $\left(\frac{1}{p}\right) = 1$ by Proposition ??(a).
- Next: compute $\left(\frac{-1}{p}\right)$ (done below).
- Next lectures: compute $\left(\frac{2}{p}\right)$ and the general reciprocity for odd primes $\left(\frac{q}{p}\right)$.

§ **Theorem 4.6 (Value of $(-1/p)$)** Let p be an odd prime. Then

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}} = \begin{cases} 1, & \text{if } p \equiv 1 \pmod{4}, \\ -1, & \text{if } p \equiv 3 \pmod{4}. \end{cases}$$

Proof. Apply Euler's criterion with $a = -1$:

$$\left(\frac{-1}{p}\right) \equiv (-1)^{\frac{p-1}{2}} \pmod{p}.$$

But the left-hand side is ± 1 and the right-hand side equals the integer $(-1)^{(p-1)/2} \in \{\pm 1\}$, hence equality holds. Writing $p = 4k + 1$ or $p = 4k + 3$ and evaluating $(-1)^{(p-1)/2}$ gives the two cases above. $\square$

§ Examples

- $\left(\frac{-1}{7}\right) = -1$ because $7 \equiv 3 \pmod{4}$.
- $\left(\frac{-1}{53}\right) = 1$ because $53 \equiv 1 \pmod{4}$.

§ Lemma 4.7 (Gauss's Lemma) Let p be an odd prime and let $a \in \mathbb{Z}$ with $p \nmid a$. For $k = 1, 2, \dots, \frac{p-1}{2}$ let r_k be the least positive residue of ka modulo p (i.e., $r_k \in \{1, \dots, p-1\}$ and $r_k \equiv ka \pmod{p}$). Let n be the number of these residues r_k that are greater than $p/2$. Then

$$\left(\frac{a}{p}\right) = (-1)^n.$$

§ Example 4.5 (Gauss's Lemma) Use Gauss's Lemma to compute $\left(\frac{6}{11}\right)$.

Consider the least positive residues of

$$6, 2 \cdot 6, 3 \cdot 6, 4 \cdot 6, 5 \cdot 6 \pmod{11}.$$

We have

$$6 \equiv 6, \quad 12 \equiv 1, \quad 18 \equiv 7, \quad 24 \equiv 2, \quad 30 \equiv 8 \pmod{11},$$

so the least positive residues are $\{6, 1, 7, 2, 8\}$. Those greater than $11/2 = 5.5$ are $6, 7, 8$, so $n = 3$. Hence

$$\left(\frac{6}{11}\right) = (-1)^3 = -1.$$

0.13 L13: 4.2 Legendre symbol, 4.3 Quadratic reciprocity

L13: Gauss' Lemma (continued) and Consequences

§ Proof of Lemma 4.7 (Gauss's Lemma)

Proof. Let p be an odd prime and $a \in \mathbb{Z}$ with $p \nmid a$. Recall the notation from the lemma: for $k = 1, 2, \dots, \frac{p-1}{2}$ let r_k denote the least positive residue of ka modulo p . Partition these residues into two classes:

$$\{r_1, \dots, r_n\} \quad (\text{those } r_k > p/2) \quad \text{and} \quad \{s_1, \dots, s_m\} \quad (\text{those } s_k < p/2),$$

so $n + m = (p-1)/2$. (No least positive residue equals $p/2$ because p is odd; none equals 0 because $p \nmid a$.)

Consider the $(p-1)/2$ integers

$$p - r_1, p - r_2, \dots, p - r_n, s_1, s_2, \dots, s_m. \quad (0.1)$$

Each integer in (??) lies in the range $1, \dots, (p-1)/2$. We claim that the integers in (??) are pairwise distinct modulo p , hence they form a permutation of $\{1, 2, \dots, (p-1)/2\}$.

First, no two of the numbers $p - r_i$ can be congruent modulo p , since $p - r_i \equiv p - r_j \pmod{p}$ would imply $r_i \equiv r_j \pmod{p}$, contradicting the uniqueness of least residues $r_i \neq r_j$ coming from distinct multipliers $k_i \neq k_j$ (with $1 \leq k_i, k_j \leq (p-1)/2$).

Second, no two of the s_j are congruent modulo p for the same reason: they are distinct least residues.

Third, no $p - r_i$ is congruent to any s_j modulo p : otherwise $p - r_i \equiv s_j \pmod{p}$, hence $r_i + s_j \equiv 0 \pmod{p}$, so $r_i \equiv -s_j \pmod{p}$. But r_i arises from $k_i a$ and s_j from $k_j a$ for some $k_i \neq k_j$ in $\{1, \dots, (p-1)/2\}$, so this would give

$$k_i a \equiv -k_j a \pmod{p} \implies (k_i + k_j)a \equiv 0 \pmod{p}.$$

Since $p \nmid a$, this forces $k_i + k_j \equiv 0 \pmod{p}$. But $1 \leq k_i, k_j \leq (p-1)/2$ implies $1 \leq k_i + k_j \leq p-1$, so the only way $k_i + k_j \equiv 0 \pmod{p}$ is $k_i + k_j = p$, which contradicts $k_i, k_j \leq (p-1)/2$. Thus no such congruence can occur.

Therefore the list (??) is a permutation of $1, 2, \dots, (p-1)/2$. Taking the product of all entries of (??) gives

$$\prod_{i=1}^n (p - r_i) \cdot \prod_{j=1}^m s_j \equiv \left(\frac{p-1}{2}\right)! \pmod{p}.$$

Since $p - r_i \equiv -r_i \pmod{p}$, this becomes

$$(-1)^n \left(\prod_{i=1}^n r_i\right) \left(\prod_{j=1}^m s_j\right) \equiv \left(\frac{p-1}{2}\right)! \pmod{p}.$$

By construction the multiset $\{r_1, \dots, r_n, s_1, \dots, s_m\}$ coincides with the least positive residues of

$$a, 2a, \dots, \left(\frac{p-1}{2}\right)a,$$

so

$$\left(\prod_{i=1}^n r_i\right) \left(\prod_{j=1}^m s_j\right) \equiv a \cdot 2a \cdots \left(\frac{p-1}{2}a\right) = a^{\frac{p-1}{2}} \left(\frac{p-1}{2}\right)! \pmod{p}.$$

Combining the last two displayed congruences,

$$(-1)^n a^{\frac{p-1}{2}} \left(\frac{p-1}{2}\right)! \equiv \left(\frac{p-1}{2}\right)! \pmod{p}.$$

Since $\gcd((\frac{p-1}{2})!, p) = 1$, we may cancel the factorial modulo p to obtain

$$(-1)^n a^{\frac{p-1}{2}} \equiv 1 \pmod{p},$$

so

$$a^{\frac{p-1}{2}} \equiv (-1)^n \pmod{p}.$$

By Euler's criterion (Theorem ??) the left-hand side equals (a/p) , which takes the value ± 1 , giving

$$\left(\frac{a}{p}\right) = (-1)^n.$$

This completes the proof of Gauss's Lemma. □

§ Theorem 4.8 (Value of $(2/p)$) Let p be an odd prime. Then

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}} = \begin{cases} 1, & \text{if } p \equiv 1, 7 \pmod{8}, \\ -1, & \text{if } p \equiv 3, 5 \pmod{8}. \end{cases}$$

Proof. Apply Gauss's Lemma (Lemma ??): let n denote the number of least positive residues of

$$2, 2 \cdot 2, 3 \cdot 2, \dots, \left(\frac{p-1}{2}\right) 2$$

that are greater than $p/2$. Then $\left(\frac{2}{p}\right) = (-1)^n$.

For $1 \leq k \leq (p-1)/2$ the product $k \cdot 2$ satisfies $k \cdot 2 < p/2$ iff $k < p/4$. Hence exactly $\lfloor \frac{p}{4} \rfloor$ of the values $k \cdot 2$ produce residues $< p/2$, and thus

$$n = \frac{p-1}{2} - \left\lfloor \frac{p}{4} \right\rfloor.$$

We need the parity of n . We claim

$$\frac{p-1}{2} - \left\lfloor \frac{p}{4} \right\rfloor \equiv \frac{p^2-1}{8} \pmod{2}.$$

It suffices to check this congruence for the four residue classes of p modulo 8.

Case $p \equiv 1 \pmod{8}$: write $p = 8k + 1$. Then

$$\frac{p-1}{2} - \left\lfloor \frac{p}{4} \right\rfloor = 4k - \lfloor 2k + \frac{1}{4} \rfloor = 4k - 2k = 2k \equiv 0 \pmod{2},$$

and

$$\frac{p^2-1}{8} = \frac{(8k+1)^2-1}{8} = 8k^2 + 2k \equiv 0 \pmod{2}.$$

Case $p \equiv 3 \pmod{8}$: write $p = 8k + 3$. Then

$$\frac{p-1}{2} - \left\lfloor \frac{p}{4} \right\rfloor = 4k + 1 - \lfloor 2k + \frac{3}{4} \rfloor = 4k + 1 - (2k + 0) = 2k + 1 \equiv 1 \pmod{2},$$

and

$$\frac{p^2-1}{8} = \frac{(8k+3)^2-1}{8} = 8k^2 + 6k + 1 \equiv 1 \pmod{2}.$$

Case $p \equiv 5 \pmod{8}$: write $p = 8k + 5$. Then

$$\frac{p-1}{2} - \left\lfloor \frac{p}{4} \right\rfloor = 4k + 2 - \lfloor 2k + 1 + \frac{1}{4} \rfloor = 4k + 2 - (2k + 1) = 2k + 1 \equiv 1 \pmod{2},$$

and

$$\frac{p^2 - 1}{8} = \frac{(8k + 5)^2 - 1}{8} = 8k^2 + 10k + 3 \equiv 1 \pmod{2}.$$

Case $p \equiv 7 \pmod{8}$: write $p = 8k + 7$. Then

$$\frac{p-1}{2} - \left\lfloor \frac{p}{4} \right\rfloor = 4k + 3 - \lfloor 2k + 1 + \frac{3}{4} \rfloor = 4k + 3 - (2k + 1) = 2k + 2 \equiv 0 \pmod{2},$$

and

$$\frac{p^2 - 1}{8} = \frac{(8k + 7)^2 - 1}{8} = 8k^2 + 14k + 6 \equiv 0 \pmod{2}.$$

Thus in all cases the desired parity congruence holds, and therefore

$$\left(\frac{2}{p} \right) = (-1)^{\frac{p^2-1}{8}},$$

which yields the stated residue-class characterization. □

§ Examples

- $\left(\frac{2}{7} \right) = 1$ since $7 \equiv 7 \pmod{8}$.
- $\left(\frac{2}{53} \right) = -1$ since $53 \equiv 5 \pmod{8}$.

§ Quadratic Reciprocity — statement The law of quadratic reciprocity relates $\left(\frac{q}{p} \right)$ and $\left(\frac{p}{q} \right)$ for odd primes $p \neq q$, often simplifying computations of Legendre symbols.

§ Theorem 4.9 (Quadratic Reciprocity) Let p and q be distinct odd primes. Then

$$\left(\frac{p}{q} \right) \left(\frac{q}{p} \right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}.$$

Equivalently,

$$\left(\frac{p}{q} \right) = \left(\frac{q}{p} \right) \quad \text{unless } p \equiv q \equiv 3 \pmod{4},$$

in which case

$$\left(\frac{p}{q} \right) = - \left(\frac{q}{p} \right).$$

§ Remark If $p \equiv 1 \pmod{4}$ or $q \equiv 1 \pmod{4}$, then the sign is $+1$ and $(p/q) = (q/p)$. If $p \equiv q \equiv 3 \pmod{4}$, the sign is -1 and $(p/q) = -(q/p)$.

§ Example 4.6: Compute $(7/53)$ Since $53 \equiv 1 \pmod{4}$, by Quadratic Reciprocity (Theorem ??) we have

$$\left(\frac{7}{53} \right) = \left(\frac{53}{7} \right).$$

Reduce 53 modulo 7: $53 \equiv 4 \pmod{7}$. Thus

$$\left(\frac{7}{53} \right) = \left(\frac{4}{7} \right) = 1,$$

because $4 \equiv 2^2$ is a square modulo 7.

§ Example 4.7: Compute $\left(\frac{-158}{101}\right)$ We compute

$$\left(\frac{-158}{101}\right) = \left(\frac{-1}{101}\right) \left(\frac{158}{101}\right).$$

Since $101 \equiv 1 \pmod{4}$, $\left(\frac{-1}{101}\right) = 1$, so

$$\left(\frac{-158}{101}\right) = \left(\frac{158}{101}\right).$$

Reduce 158 (mod 101): $158 \equiv 57 \pmod{101}$. Factor $57 = 3 \cdot 19$, so multiplicativity gives

$$\left(\frac{57}{101}\right) = \left(\frac{3}{101}\right) \left(\frac{19}{101}\right).$$

Compute $\left(\frac{3}{101}\right)$ by reciprocity. Since $(3-1)/2 = 1$ and $(101-1)/2 = 50$, their product is even, so no sign change occurs:

$$\left(\frac{3}{101}\right) = \left(\frac{101}{3}\right) = \left(\frac{2}{3}\right) = -1,$$

because 2 is a nonresidue modulo 3.

Next compute $\left(\frac{19}{101}\right)$. Here

$$\left(\frac{19}{101}\right) = \left(\frac{101}{19}\right)$$

since $((19-1)/2)((101-1)/2) = 9 \cdot 50$ is even. Reduce 101 (mod 19): $101 \equiv 6 \pmod{19}$, hence

$$\left(\frac{19}{101}\right) = \left(\frac{6}{19}\right) = \left(\frac{2}{19}\right) \left(\frac{3}{19}\right).$$

Using Theorem ??, $\left(\frac{2}{19}\right) = -1$ because $19 \equiv 3 \pmod{8}$. Also

$$\left(\frac{3}{19}\right) = -\left(\frac{19}{3}\right)$$

by reciprocity (the sign is -1 because $(3-1)/2 \cdot (19-1)/2 = 1 \cdot 9$ is odd). Now $19 \equiv 1 \pmod{3}$, so $\left(\frac{19}{3}\right) = \left(\frac{1}{3}\right) = 1$, hence $\left(\frac{3}{19}\right) = -1$. Therefore $\left(\frac{6}{19}\right) = (-1) \cdot (-1) = 1$, so $\left(\frac{19}{101}\right) = 1$.

Combining the two factors,

$$\left(\frac{57}{101}\right) = \left(\frac{3}{101}\right) \left(\frac{19}{101}\right) = (-1) \cdot 1 = -1,$$

and hence

$$\left(\frac{-158}{101}\right) = -1.$$

§ Warning Do not apply Quadratic Reciprocity to composite moduli. For example, it is incorrect to write

$$\left(\frac{57}{101}\right) = \left(\frac{101}{57}\right)$$

because 57 is not prime. Quadratic Reciprocity applies only to odd prime moduli.

0.14 L14: 4.3 Proof quadratic reciprocity and examples

L14: Proof of the Quadratic Reciprocity Law (Theorem 4.9)

§ Lemma 4.10 Let p be an odd prime and let $a \in \mathbb{Z}$ with $p \nmid a$ and a odd. Define

$$N := \sum_{j=1}^{\frac{p-1}{2}} \left\lfloor \frac{ja}{p} \right\rfloor.$$

Then

$$\left(\frac{a}{p} \right) = (-1)^N.$$

Here $\lfloor \cdot \rfloor$ denotes the floor function.

§ Remark (role of Lemma 4.10) Lemma ??, like Euler's criterion and Gauss's Lemma, is often more useful theoretically than computationally; nevertheless it provides an alternate viewpoint that is convenient for proving reciprocity.

§ Example 4.8 (Use Lemma 4.10 to compute $(7/11)$) Compute $(\frac{7}{11})$ using Lemma ??. For $j = 1, \dots, 5$ compute

$$\left\lfloor \frac{7}{11} \right\rfloor = 0, \quad \left\lfloor \frac{14}{11} \right\rfloor = 1, \quad \left\lfloor \frac{21}{11} \right\rfloor = 1, \quad \left\lfloor \frac{28}{11} \right\rfloor = 2, \quad \left\lfloor \frac{35}{11} \right\rfloor = 3,$$

so

$$N = 0 + 1 + 1 + 2 + 3 = 7.$$

Therefore $(\frac{7}{11}) = (-1)^7 = -1$.

§ Proof of Lemma 4.10

Proof. Let p and a satisfy the hypotheses. For $k = 1, \dots, \frac{p-1}{2}$ write the division identity

$$ka = p \left\lfloor \frac{ka}{p} \right\rfloor + r_k, \tag{0.2}$$

where r_k is the remainder, $0 \leq r_k \leq p-1$. Since $p \nmid a$, none of the remainders equals 0. Partition the least positive remainders r_k into

$$\{r_1, \dots, r_n\} \quad (\text{those } r_k > p/2) \quad \text{and} \quad \{s_1, \dots, s_m\} \quad (\text{those } s_k < p/2),$$

so $n + m = (p-1)/2$.

Summing (??) over $k = 1, \dots, (p-1)/2$ yields

$$\sum_{k=1}^{\frac{p-1}{2}} ka = p \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{ka}{p} \right\rfloor + \sum_{k=1}^{\frac{p-1}{2}} r_k. \tag{2}$$

From the Gauss Lemma proof (cf. Lemma ??), the multiset of remainders $\{r_1, \dots, r_n, s_1, \dots, s_m\}$ coincides with the least positive residues of $a, 2a, \dots, \frac{p-1}{2}a$. Moreover, the integers

$$p - r_1, \dots, p - r_n, s_1, \dots, s_m$$

form a permutation of $1, 2, \dots, \frac{p-1}{2}$. Hence

$$\sum_{j=1}^{\frac{p-1}{2}} j = \sum_{i=1}^n (p - r_i) + \sum_{j=1}^m s_j = pn - \sum_{i=1}^n r_i + \sum_{j=1}^m s_j. \quad (3)$$

But $\sum_{k=1}^{(p-1)/2} r_k = \sum_{i=1}^n r_i + \sum_{j=1}^m s_j$. Subtracting (3) from (2) gives

$$\sum_{k=1}^{\frac{p-1}{2}} ka - \sum_{j=1}^{\frac{p-1}{2}} j = p \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{ka}{p} \right\rfloor - pn + 2 \sum_{i=1}^n r_i. \quad (4)$$

Reduce both sides of (4) modulo 2. Note that $\sum_{k=1}^{(p-1)/2} ka - \sum_{j=1}^{(p-1)/2} j$ is an integer multiple of $(p-1)/2$ and hence even or odd depending on $(p-1)/2$, but the parity analysis below yields

$$0 \equiv p \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{ka}{p} \right\rfloor - pn \pmod{2}.$$

Since p is odd we may cancel p modulo 2 to obtain

$$0 \equiv \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{ka}{p} \right\rfloor - n \pmod{2},$$

so

$$n \equiv \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{ka}{p} \right\rfloor \equiv N \pmod{2}.$$

By Gauss's Lemma (Lemma ??) we know $\left(\frac{a}{p}\right) = (-1)^n$, hence

$$\left(\frac{a}{p}\right) = (-1)^n = (-1)^N,$$

which proves the lemma. □

§ Proof of Theorem 4.9 (Quadratic Reciprocity)

Proof. We prove the formula

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}$$

for distinct odd primes p and q . Without loss of generality assume $p > q$.

Consider the rectangle in the first quadrant with corner coordinates $(0, 0)$, $(\frac{p-1}{2}, 0)$, $(\frac{p-1}{2}, \frac{q-1}{2})$, $(0, \frac{q-1}{2})$ and count lattice points with integer coordinates (x, y) satisfying $1 \leq x \leq \frac{p-1}{2}$ and $1 \leq y \leq \frac{q-1}{2}$. The total number of such lattice points is

$$\frac{p-1}{2} \cdot \frac{q-1}{2}.$$

We now count these lattice points by grouping them according to whether they lie below or above the line $y = \frac{q}{p}x$ (the diagonal ON in the textbook figure).

Let

$$N_1 := \sum_{j=1}^{\frac{q-1}{2}} \left\lfloor \frac{jp}{q} \right\rfloor \quad \text{and} \quad N_2 := \sum_{j=1}^{\frac{p-1}{2}} \left\lfloor \frac{jq}{p} \right\rfloor.$$

A standard lattice-point argument (see Strayer's Figure 4.1) shows that N_1 counts the integer lattice points strictly below the line in one triangular region and N_2 counts the points in the complementary triangular region, and that every lattice point in the rectangle is counted exactly once by $N_1 + N_2$. Therefore

$$N_1 + N_2 = \frac{p-1}{2} \cdot \frac{q-1}{2}. \quad (5)$$

By Lemma ?? (applied separately to the pairs (p, q) and (q, p)) we have

$$\left(\frac{p}{q}\right) = (-1)^{N_1}, \quad \left(\frac{q}{p}\right) = (-1)^{N_2}.$$

Hence

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{N_1+N_2} = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}},$$

where we used (??). This is exactly the statement of Quadratic Reciprocity. □

§ Example 4.9 (Characterize primes p for which 11 is a quadratic residue) We seek all odd primes $p \neq 11$ such that $\left(\frac{11}{p}\right) = 1$.

By Quadratic Reciprocity (Theorem ??),

$$\left(\frac{11}{p}\right) = \begin{cases} \left(\frac{p}{11}\right), & \text{if } p \equiv 1 \pmod{4}, \\ -\left(\frac{p}{11}\right), & \text{if } p \equiv 3 \pmod{4}. \end{cases}$$

The quadratic residues modulo 11 (nonzero squares) are 1, 3, 4, 5, 9 and non-residues are 2, 6, 7, 8, 10.

Thus for $p \equiv 1 \pmod{4}$ we need $p \pmod{11} \in \{1, 3, 4, 5, 9\}$, and for $p \equiv 3 \pmod{4}$ we need $p \pmod{11} \in \{2, 6, 7, 8, 10\}$. Solving these congruence pairs modulo 44 (via the Chinese Remainder Theorem) yields the residue classes modulo 44 of primes p for which $\left(\frac{11}{p}\right) = 1$:

$$p \equiv 1, 5, 9, 25, 37 \pmod{44} \quad \text{or} \quad p \equiv 7, 19, 35, 39, 43 \pmod{44}.$$

Equivalently, the ten congruence classes modulo 44 are

$$1, 5, 7, 9, 19, 25, 35, 37, 39, 43 \pmod{44}.$$

§ Exercise / Further example Characterize the primes p for which a given integer s is a quadratic residue modulo p (use factorization, multiplicativity and Quadratic Reciprocity).

0.15 L15: 5.1 Primitive roots and order of an integer modulo m .

L15: Primitive Roots

§ Definition 5.0 (Order of an integer modulo m) Let $m > 0$ be an integer and let $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. The *order* of a modulo m , denoted $\text{ord}_m(a)$, is the least positive integer n such that

$$a^n \equiv 1 \pmod{m}.$$

§ Remark By Euler's theorem (Theorem ??), $a^{\varphi(m)} \equiv 1 \pmod{m}$, so the order $\text{ord}_m(a)$ exists and satisfies

$$\text{ord}_m(a) \mid \varphi(m).$$

Thus the order is always a divisor of $\varphi(m)$.

§ Examples (orders modulo 7 and 13)

- Compute $\text{ord}_7(2)$. One checks

$$2^1 \equiv 2, 2^2 \equiv 4, 2^3 \equiv 8 \equiv 1 \pmod{7},$$

so $\text{ord}_7(2) = 3$.

- Compute $\text{ord}_7(3)$. One checks

$$3^1 \equiv 3, 3^2 \equiv 2, 3^3 \equiv 6, 3^4 \equiv 4, 3^5 \equiv 5, 3^6 \equiv 1 \pmod{7},$$

so $\text{ord}_7(3) = 6 = \varphi(7)$.

- Compute $\text{ord}_{13}(2)$. By Proposition ?? below the order divides $\varphi(13) = 12$. We compute

$$2^1 \equiv 2, 2^2 \equiv 4, 2^3 \equiv 8, 2^4 \equiv 16 \equiv 3, 2^6 \equiv (2^3)^2 \equiv 8^2 \equiv 64 \equiv 12 \equiv -1 \pmod{13}.$$

Since $2^6 \equiv -1 \not\equiv 1$, the order is not 1, 2, 3, or 6, hence must be 12. Thus $\text{ord}_{13}(2) = 12$.

§ Proposition 5.1 Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. For a positive integer n we have

$$a^n \equiv 1 \pmod{m} \iff \text{ord}_m(a) \mid n.$$

In particular $\text{ord}_m(a) \mid \varphi(m)$.

Proof. ($\Rightarrow$) Suppose $a^n \equiv 1 \pmod{m}$. Write $n = q \text{ord}_m(a) + r$ with integers $q \geq 0$ and $0 \leq r < \text{ord}_m(a)$ by the Division Algorithm. Then

$$a^n = a^{q \text{ord}_m(a) + r} = (a^{\text{ord}_m(a)})^q a^r \equiv a^r \pmod{m}.$$

Since $a^n \equiv 1 \pmod{m}$, it follows that $a^r \equiv 1 \pmod{m}$. By minimality of $\text{ord}_m(a)$ we must have $r = 0$. Hence $n = q \text{ord}_m(a)$ and $\text{ord}_m(a) \mid n$.

($\Leftarrow$) If $\text{ord}_m(a) \mid n$ then $n = \text{ord}_m(a) b$ for some $b \in \mathbb{Z}_{>0}$, so

$$a^n = (a^{\text{ord}_m(a)})^b \equiv 1^b \equiv 1 \pmod{m}.$$

This completes the equivalence. The last statement follows from Euler's theorem $a^{\varphi(m)} \equiv 1 \pmod{m}$, hence $\text{ord}_m(a) \mid \varphi(m)$ by the first part. $\square$

§ Proposition 5.2 Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. For nonnegative integers i, j we have

$$a^i \equiv a^j \pmod{m} \iff i \equiv j \pmod{\text{ord}_m(a)}.$$

Proof. Assume without loss of generality $i \geq j$.

($\Rightarrow$) If $a^i \equiv a^j \pmod{m}$, then $a^{i-j} \equiv 1 \pmod{m}$. By Proposition ?? we have $\text{ord}_m(a) \mid (i - j)$, so $i \equiv j \pmod{\text{ord}_m(a)}$.

($\Leftarrow$) If $i \equiv j \pmod{\text{ord}_m(a)}$ then $i - j = t \text{ord}_m(a)$ for some t , and hence

$$a^i = a^{j+t \text{ord}_m(a)} = a^j (a^{\text{ord}_m(a)})^t \equiv a^j \cdot 1^t \equiv a^j \pmod{m}.$$

This proves the equivalence. □

§ Definition 5.1 (Primitive root modulo m) Let $m > 0$. An integer r with $\gcd(r, m) = 1$ is called a *primitive root modulo m* if

$$\text{ord}_m(r) = \varphi(m).$$

§ Examples (primitive roots)

- 3 is a primitive root modulo 7 since $\text{ord}_7(3) = 6 = \varphi(7)$.
- 2 is a primitive root modulo 13 since $\text{ord}_{13}(2) = 12 = \varphi(13)$ (see above).
- 2 is *not* a primitive root modulo 7 because $\text{ord}_7(2) = 3 \neq 6$.

§ Nonexistence example (modulo 8) The reduced residue system modulo 8 is $\{1, 3, 5, 7\}$ and $\varphi(8) = 4$. One checks

$$\text{ord}_8(1) = 1, \quad \text{ord}_8(3) = 2, \quad \text{ord}_8(5) = 2, \quad \text{ord}_8(7) = 2,$$

so no element has order 4. Therefore there are no primitive roots modulo 8.

§ Proposition 5.3 If r is a primitive root modulo m , then the set

$$\{r, r^2, r^3, \dots, r^{\varphi(m)}\}$$

is a reduced residue system modulo m (i.e., a complete set of representatives of the invertible residue classes mod m).

Proof. Since $\gcd(r, m) = 1$, each r^k is coprime to m . There are $\varphi(m)$ elements in the set; to show they are a reduced residue system it suffices to show they are pairwise incongruent modulo m . If $r^i \equiv r^j \pmod{m}$ with $1 \leq i < j \leq \varphi(m)$ then Proposition ?? implies $i \equiv j \pmod{\text{ord}_m(r)}$. But $\text{ord}_m(r) = \varphi(m)$, so $0 < j - i < \varphi(m)$ cannot be a multiple of $\varphi(m)$; hence no such congruence can occur and the elements are pairwise distinct modulo m . Thus the set is a reduced residue system. □

§ Remark If a primitive root modulo m exists, it is typically not unique: if r is a primitive root and $\gcd(k, \varphi(m)) = 1$, then r^k is also a primitive root modulo m . In a later lecture we will determine precisely how many primitive roots modulo m exist when they exist at all.

§ Exercises

1. Show there are no primitive roots modulo 12.
2. Prove that if r is a primitive root modulo m , then exactly $\varphi(\varphi(m))$ of the elements r^k ($1 \leq k \leq \varphi(m)$) are primitive

roots modulo m (hint: count k with $\gcd(k, \varphi(m)) = 1$).

0.16 L16: 5.1 Primitive roots and order of an integer modulo m . & 5.2 Primitive roots for prime moduli.

L16: The order of an integer (continued)

§ Proposition 5.4 Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. For any positive integer i ,

$$\text{ord}_m(a^i) = \frac{\text{ord}_m(a)}{\gcd(\text{ord}_m(a), i)}.$$

§ Proof

Proof. Write $t := \text{ord}_m(a)$ and set $d = \gcd(t, i)$. Then there exist coprime integers $b, c > 0$ with

$$t = db, \quad i = dc, \quad \gcd(b, c) = 1.$$

Compute

$$(a^i)^b = a^{ib} = a^{(dc)(b)} = a^{tc} = (a^t)^c \equiv 1 \pmod{m},$$

so by Proposition ?? we have $\text{ord}_m(a^i) \mid b$.

Conversely, since $a^t \equiv 1 \pmod{m}$ we have

$$a^{i \text{ord}_m(a^i)} \equiv 1 \pmod{m},$$

so by Proposition ?? it follows that $t \mid i \text{ord}_m(a^i)$. Substituting $t = db$ and $i = dc$ gives

$$db \mid dc \cdot \text{ord}_m(a^i) \implies b \mid c \cdot \text{ord}_m(a^i).$$

Because $\gcd(b, c) = 1$, this implies $b \mid \text{ord}_m(a^i)$. Combining with the previous divisibility $\text{ord}_m(a^i) \mid b$, we get

$$\text{ord}_m(a^i) = b = \frac{t}{d} = \frac{\text{ord}_m(a)}{\gcd(\text{ord}_m(a), i)}.$$

This completes the proof. □

§ Corollary 5.5 Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. For a positive integer i ,

$$\text{ord}_m(a^i) = \text{ord}_m(a) \iff \gcd(\text{ord}_m(a), i) = 1.$$

§ Proof

Proof. Immediate from Proposition ??: $\text{ord}_m(a^i) = \text{ord}_m(a) / \gcd(\text{ord}_m(a), i)$ equals $\text{ord}_m(a)$ iff the denominator is 1. □

§ Corollary 5.6 (Counting primitive roots) If a primitive root modulo m exists, then there are exactly $\varphi(\varphi(m))$ incongruent primitive roots modulo m .

§ Proof

Proof. Let r be a primitive root modulo m , so $\text{ord}_m(r) = \varphi(m)$. By Proposition ??, the powers

$$r, r^2, \dots, r^{\varphi(m)}$$

form a reduced-residue system modulo m and hence represent all $\varphi(m)$ residue classes coprime to m . By Corollary ??, r^i is a primitive root (i.e. has order $\varphi(m)$) if and only if $\gcd(\varphi(m), i) = 1$. The number of integers i in $\{1, \dots, \varphi(m)\}$ satisfying this is $\varphi(\varphi(m))$. Thus there are exactly $\varphi(\varphi(m))$ incongruent primitive roots modulo m . $\square$

§ Example 5.1 We know 3 is a primitive root modulo 7. Since $\varphi(7) = 6$, Corollary ?? predicts $\varphi(\varphi(7)) = \varphi(6) = 2$ primitive roots modulo 7. Indeed, $3^1 \equiv 3$ and $3^5 \equiv 5 \pmod{7}$ are the two incongruent primitive roots modulo 7.

§ Remark Corollary ?? applies only when at least one primitive root modulo m exists. For example, although $\varphi(\varphi(8)) = \varphi(4) = 2$, there are no primitive roots modulo 8 (see previous lecture), so the corollary does not apply in that case.

§ Section: Primitive roots for prime moduli

§ Theorem 5.7 (Lagrange's Theorem for polynomials over $\mathbb{F}_p$) Let p be a prime and let

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

be a polynomial with integer coefficients of degree $n \geq 1$, and assume $p \nmid a_n$. Then the congruence

$$f(x) \equiv 0 \pmod{p}$$

has at most n incongruent solutions modulo p .

§ Proof

Proof. We work in the field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. Proceed by induction on n . The base case $n = 1$ is elementary: a nonzero linear polynomial has exactly one root in $\mathbb{F}_p$.

Assume the statement holds for degree $\leq k$ and let f have degree $k + 1$. If f has no root then the conclusion is trivial. Otherwise let $a \in \mathbb{F}_p$ be a root. Perform polynomial division in $\mathbb{F}_p[x]$:

$$f(x) = (x - a)q(x) + r,$$

with $q(x) \in \mathbb{F}_p[x]$ of degree k and $r \in \mathbb{F}_p$. Evaluating at $x = a$ gives $0 = f(a) = r$, so $f(x) = (x - a)q(x)$ in $\mathbb{F}_p[x]$. Any other root $b \not\equiv a$ must satisfy $q(b) = 0$, and by the induction hypothesis q has at most k roots. Therefore f has at most $k + 1$ roots, completing the induction. $\square$

§ Proposition 5.8 Let p be a prime and let $d > 0$ be a divisor of $p - 1$. Then the congruence

$$x^d \equiv 1 \pmod{p}$$

has exactly d incongruent solutions modulo p .

§ Proof

Proof. Write $p - 1 = de$ for some integer $e \geq 1$. Factor the polynomial identity

$$x^{p-1} - 1 = x^{de} - 1 = (x^d - 1)(x^{d(e-1)} + \dots + x^d + 1).$$

By Fermat's little theorem the congruence $x^{p-1} \equiv 1 \pmod{p}$ holds for each $x \in \{1, 2, \dots, p - 1\}$, so there are $p - 1$ solutions among the nonzero residues. Each such solution is a solution of either

$$x^d \equiv 1 \pmod{p} \quad \text{or} \quad x^{d(e-1)} + \dots + x^d + 1 \equiv 0 \pmod{p}.$$

The second polynomial has degree $d(e - 1) = p - 1 - d$, hence by Theorem ?? it has at most $p - 1 - d$ solutions. Thus the congruence $x^d \equiv 1 \pmod{p}$ must have at least

$$(p - 1) - (p - 1 - d) = d$$

solutions. But Theorem ?? also implies $x^d - 1$ (a degree- d polynomial) has at most d roots modulo p . Combining the two bounds yields exactly d distinct solutions modulo p , as required. $\square$

§ Example / Exercise

- Exercise: Assuming 17 has a primitive root, determine how many incongruent primitive roots there are modulo 17. (*Hint*: use Corollary ??.)
- Do the same for modulus 18 (be careful: check whether primitive roots modulo 18 exist before applying Corollary ??).

§ Further exercises

- Prove that 3 is a primitive root modulo 43.
- Using the previous part, find all incongruent integers of order 14 modulo 43.

0.17 L17: 5.2 Primitive roots for prime moduli & 5.3 The Primitive root Theorem.

L17: Primitive Roots for Prime Numbers

§ Theorem 5.9 (Legendre) Let p be a prime and let d be a positive divisor of $p - 1$ (i.e. $d \mid (p - 1)$). Then the number of incongruent integers modulo p whose order equals d is $\varphi(d)$.

§ Proof

Proof. For each positive divisor $d \mid (p - 1)$ let $f(d)$ denote the number of integers a with $1 \leq a \leq p - 1$ such that $\text{ord}_p(a) = d$. By Proposition ?? every nonzero residue mod p has order dividing $p - 1$, hence

$$\sum_{d \mid (p-1)} f(d) = p - 1.$$

On the other hand, by elementary properties of the totient function,

$$\sum_{d \mid (p-1)} \varphi(d) = p - 1.$$

Thus

$$\sum_{d \mid (p-1)} f(d) = \sum_{d \mid (p-1)} \varphi(d). \quad (0.3)$$

To conclude $f(d) = \varphi(d)$ for every $d \mid (p - 1)$ it suffices to show $f(d) \leq \varphi(d)$ for each such d . If $f(d) = 0$ this is trivial, so assume $f(d) > 0$. Then there exists a with $\text{ord}_p(a) = d$. The elements

$$a, a^2, \dots, a^d$$

are pairwise incongruent modulo p (if $a^i \equiv a^j$ with $0 \leq i < j \leq d$ then $a^{j-i} \equiv 1$ with $1 \leq j - i < d$, contradicting minimality of d). Hence they are d distinct solutions of

$$x^d \equiv 1 \pmod{p}.$$

By Proposition ?? the congruence $x^d \equiv 1 \pmod{p}$ has exactly d incongruent solutions; therefore the full solution set is precisely $\{a, a^2, \dots, a^d\}$, which forms a cyclic subgroup of order d . Among these d solutions exactly $\varphi(d)$ are generators of the cyclic subgroup (i.e. have order d) — that is, exactly the integers a^k with $\gcd(k, d) = 1$ have order d . Therefore $f(d) = \varphi(d)$, as required. $\square$

§ Corollary 5.10 Let p be a prime. Then the number of incongruent primitive roots modulo p is $\varphi(p - 1)$. In particular, primitive roots modulo p exist for every prime p .

§ Proof

Proof. A primitive root modulo p is precisely an integer of order $p - 1$. By Theorem ??, the number of such incongruent integers is $\varphi(p - 1)$. In particular $\varphi(p - 1) > 0$, so primitive roots exist modulo p . $\square$

§ Example 5.2 (Orders modulo 7) Theorem ?? guarantees existence of integers of orders dividing $\varphi(7) = 6$, namely orders 1, 2, 3, 6. The count and classes are:

order d	# of residue classes	residue classes (mod 7)
1	$\varphi(1) = 1$	$\{1\}$
2	$\varphi(2) = 1$	$\{6\}$
3	$\varphi(3) = 2$	$\{2, 4\}$
6	$\varphi(6) = 2$	$\{3, 5\}$

§ Example 5.3 (Find integers of order 6 modulo 19) By Corollary ?? there are exactly $\varphi(6) = 2$ incongruent integers of order 6 modulo 19. One convenient approach is to find a primitive root modulo 19 and then use Proposition ??.

First show 2 is a primitive root modulo 19. By Proposition ?? it suffices to check $2^d \not\equiv 1 \pmod{19}$ for each proper divisor d of $18 = 2 \cdot 3^2$, i.e. $d \in \{1, 2, 3, 6, 9\}$. Computations give

$$2^1 \equiv 2, 2^2 \equiv 4, 2^3 \equiv 8, 2^6 \equiv 7, 2^9 \equiv 18 \pmod{19},$$

none of which are 1, so $\text{ord}_{19}(2) = 18$. Thus 2 is primitive.

By Proposition ??,

$$\text{ord}_{19}(2^i) = \frac{18}{\gcd(18, i)}.$$

We seek i with $\text{ord}_{19}(2^i) = 6$, i.e. $\frac{18}{\gcd(18, i)} = 6$, so $\gcd(18, i) = 3$. For $1 \leq i \leq 18$ the solutions with $\gcd(18, i) = 3$ are $i = 3, 15$. Hence the elements 2^3 and 2^{15} have order 6.

Compute:

$$2^3 \equiv 8 \pmod{19}.$$

Also

$$2^{15} = 2^{12} \cdot 2^3 \equiv (2^6)^2 \cdot 8 \equiv 7^2 \cdot 8 \equiv 49 \cdot 8 \equiv 11 \cdot 8 \equiv 88 \equiv 12 \pmod{19},$$

so the two incongruent integers with order 6 modulo 19 are $\{8, 12\}$.

§ Remarks and conjectures

- Table 5.1 (in your text) lists least primitive roots for small primes; 2 frequently appears as the least primitive root for many primes.
- Conjecture: There are infinitely many primes p for which 2 is the least positive primitive root modulo p (empirical pattern; open).
- **Artin's conjecture:** If r is an integer which is not a perfect square and $r \neq -1$, then there are infinitely many primes p for which r is a primitive root modulo p . This remains open; however Heath-Brown (1986) proved that at most two integers r can fail Artin's conjecture in a certain strong sense.

§ Proposition 5.11 For integers $n \geq 3$ there are no primitive roots modulo 2^n .

§ Proof

Proof. If 2^n had a primitive root then it would be an odd integer of order $\varphi(2^n) = 2^{n-1}$. We show that every odd integer a satisfies

$$a^{2^{n-2}} \equiv 1 \pmod{2^n},$$

which implies the order divides $2^{n-2} < 2^{n-1}$, hence no primitive root exists.

Proof by induction on $n \geq 3$. Base case $n = 3$: any odd $a \equiv 1, 3, 5, 7 \pmod{8}$ satisfies $a^2 \equiv 1 \pmod{8}$.

Inductive step: suppose $a^{2^{k-2}} \equiv 1 \pmod{2^k}$ for some $k \geq 3$. Then write

$$a^{2^{k-2}} = 1 + 2^k t$$

for some integer t . Squaring both sides gives

$$a^{2^{k-1}} = (1 + 2^k t)^2 = 1 + 2^{k+1} t + 2^{2k} t^2 \equiv 1 \pmod{2^{k+1}},$$

since $2^{2k} t^2$ is divisible by 2^{k+1} when $k \geq 3$. This completes the induction and the proof. $\square$

§ Proposition 5.12 Let $m, n > 2$ be coprime integers. Then there are no primitive roots modulo mn .

§ Proof

Proof. Let a be any integer with $\gcd(a, mn) = 1$. Then $\gcd(a, m) = \gcd(a, n) = 1$. Let

$$d := \gcd(\varphi(m), \varphi(n)), \quad k := \text{lcm}(\varphi(m), \varphi(n)).$$

Since $m, n > 2$ we have $\varphi(m)$ and $\varphi(n)$ even, hence $d \geq 2$. Then

$$k = \frac{\varphi(m)\varphi(n)}{d} = \frac{\varphi(mn)}{d} \leq \frac{\varphi(mn)}{2}.$$

By Euler's theorem $a^{\varphi(m)} \equiv 1 \pmod{m}$, so raising both sides to the $\varphi(n)/d$ power yields

$$a^k \equiv 1 \pmod{m}.$$

Similarly $a^k \equiv 1 \pmod{n}$. Since $(m, n) = 1$, it follows that $a^k \equiv 1 \pmod{mn}$. Therefore the order of a modulo mn divides $k \leq \varphi(mn)/2$, so cannot equal $\varphi(mn)$. Hence no primitive root modulo mn exists. $\square$

§ Corollary 5.13 (Necessary form for moduli with primitive roots) If an integer $n > 0$ admits a primitive root then n must be of one of the forms

$$1, 2, 4, p^m, 2p^m$$

where p is an odd prime and $m \geq 1$.

§ Remark / Proof sketch The full proof of this corollary uses Propositions ?? and ?? and further elementary reduction arguments: any modulus n decomposes as $2^e \prod p_i^{e_i}$; one checks which local factors allow maximal cyclic multiplicative groups and eliminates mixed composite factors. The complete characterization (the Primitive Root Theorem) will be given in a later lecture.

§ Exercises

1. Construct the order/residue table for $p = 13$ (analogue of Example ??).
2. Find all incongruent integers of order 6 modulo 19 (exercise solved in Example ??).
3. Prove that 3 is a primitive root modulo 43, and list all incongruent integers having order 14 modulo 43.
4. Prove Corollary ??.

0.18 L18: 5.3 The Primitive Root Theorem

L18: The Primitive Root Theorem (continued)

§ Lemma 5.14 Let p be an odd prime. There exists a primitive root r modulo p such that

$$r^{p-1} \not\equiv 1 \pmod{p^2}.$$

§ Proof

Proof. Let r be any primitive root modulo p . If $r^{p-1} \not\equiv 1 \pmod{p^2}$ we are done. Otherwise assume

$$r^{p-1} \equiv 1 \pmod{p^2}.$$

Set $r' := r + p$. Since $r' \equiv r \pmod{p}$, r' is a primitive root modulo p . We will show $r'^{p-1} \not\equiv 1 \pmod{p^2}$.

By the binomial theorem,

$$(r + p)^{p-1} = \sum_{k=0}^{p-1} \binom{p-1}{k} r^{p-1-k} p^k.$$

All terms with $k \geq 2$ are divisible by p^2 , so modulo p^2 only the $k = 0$ and $k = 1$ terms contribute:

$$(r + p)^{p-1} \equiv r^{p-1} + \binom{p-1}{1} r^{p-2} p \pmod{p^2}.$$

Since $\binom{p-1}{1} = p - 1$, we get

$$(r + p)^{p-1} \equiv r^{p-1} + (p - 1)p r^{p-2} \equiv r^{p-1} - p r^{p-2} \pmod{p^2},$$

because $(p - 1)p \equiv -p \pmod{p^2}$. Using the assumption $r^{p-1} \equiv 1 \pmod{p^2}$ this becomes

$$(r + p)^{p-1} \equiv 1 - p r^{p-2} \pmod{p^2}.$$

If $(r + p)^{p-1} \equiv 1 \pmod{p^2}$ then $p r^{p-2} \equiv 0 \pmod{p^2}$, so $p \mid r^{p-2}$ and hence $p \mid r$, contradicting $\gcd(r, p) = 1$. Therefore $(r + p)^{p-1} \not\equiv 1 \pmod{p^2}$, and the lemma is proved. $\square$

§ Corollary 5.15 Let p be an odd prime and let r be a primitive root modulo p . Then either r or $r + p$ is a primitive root modulo p^2 .

§ Proof

Proof. Let $n := \text{ord}_{p^2}(r)$. By Proposition ??, $n \mid \varphi(p^2) = p(p - 1)$. Reducing modulo p shows $\text{ord}_p(r) = p - 1 \mid n$. Hence n is either $p - 1$ or $p(p - 1)$. If $n = p(p - 1)$ then r is a primitive root modulo p^2 and we are done. If $n = p - 1$, then by Lemma ?? the element $r' := r + p$ satisfies $r'^{p-1} \not\equiv 1 \pmod{p^2}$; repeating the divisibility argument above yields $\text{ord}_{p^2}(r') \in \{p - 1, p(p - 1)\}$, but it cannot be $p - 1$ by the lemma, so $\text{ord}_{p^2}(r') = p(p - 1)$ and r' is a primitive root modulo p^2 . $\square$

§ Lemma 5.16 Let p be an odd prime and let r be a primitive root modulo p with

$$r^{p-1} \not\equiv 1 \pmod{p^2}.$$

For every integer $m \geq 2$,

$$r^{p^{m-2}(p-1)} \not\equiv 1 \pmod{p^m}.$$

§ Proof

Proof. We prove the claim by induction on $m \geq 2$. The base case $m = 2$ is the hypothesis.

Assume the statement holds for $m = k \geq 2$, i.e.

$$r^{p^{k-2}(p-1)} \not\equiv 1 \pmod{p^k}.$$

Thus there exists an integer a with

$$r^{p^{k-2}(p-1)} = 1 + ap^{k-1},$$

and necessarily $p \nmid a$ (otherwise $r^{p^{k-2}(p-1)} \equiv 1 \pmod{p^k}$, contradicting the induction hypothesis).

Raise both sides to the p -th power:

$$r^{p^{k-1}(p-1)} = (1 + ap^{k-1})^p = 1 + p(ap^{k-1}) + \binom{p}{2}(ap^{k-1})^2 + \cdots + (ap^{k-1})^p.$$

All terms except the first two are divisible by p^{k+1} , hence modulo p^{k+1} we have

$$r^{p^{k-1}(p-1)} \equiv 1 + ap^k \pmod{p^{k+1}}.$$

Since $p \nmid a$, this is not congruent to 1 modulo p^{k+1} . Thus the statement holds for $m = k + 1$. By induction it is true for all $m \geq 2$. $\square$

§ Proposition 5.17 Let p be an odd prime and $m \geq 1$ a positive integer. Then there exists a primitive root modulo p^m .

§ Proof

Proof. The case $m = 1$ is Corollary ???. For $m \geq 2$, pick a primitive root r modulo p satisfying Lemma ??, and hence Lemma ?? (so that

$$r^{p^{m-2}(p-1)} \not\equiv 1 \pmod{p^m} \text{ for the given } m).$$

Let $n := \text{ord}_{p^m}(r)$. By Proposition ??, $n \mid \varphi(p^m) = p^{m-1}(p-1)$. Reducing modulo p gives $\text{ord}_p(r) = p-1 \mid n$. Thus $n = p^k(p-1)$ for some integer $0 \leq k \leq m-1$.

If $k \leq m-2$, then

$$r^{p^{m-2}(p-1)} = (r^{p^k(p-1)})^{p^{m-2-k}} \equiv 1^{p^{m-2-k}} \equiv 1 \pmod{p^m},$$

contradicting Lemma ??. Therefore $k = m-1$, and so

$$n = p^{m-1}(p-1) = \varphi(p^m),$$

i.e. r is a primitive root modulo p^m . $\square$

§ Corollary 5.18 Let p be an odd prime and $m \geq 1$. There exists a primitive root modulo $2p^m$.

§ Proof

Proof. By Proposition ??? choose a primitive root r modulo p^m . If r is even, replace it by $r + p^m$ (which is still a primitive root mod p^m and is odd); hence we may assume r is odd and therefore $\text{gcd}(r, 2p^m) = 1$.

Let $n := \text{ord}_{2p^m}(r)$. Then $r^n \equiv 1 \pmod{2p^m}$ implies in particular $r^n \equiv 1 \pmod{p^m}$, so $\varphi(p^m) = p^{m-1}(p-1) \mid n$ by Proposition ??. On the other hand $n \mid \varphi(2p^m) = \varphi(2)\varphi(p^m) = \varphi(p^m)$, hence $n = \varphi(p^m) = \varphi(2p^m)$ and r is a primitive root modulo $2p^m$. $\square$

§ Theorem 5.19 (Primitive Root Theorem) Let $n \geq 1$ be an integer. There exists a primitive root modulo n if and only if

$$n \in \{1, 2, 4, p^m, 2p^m\},$$

where p is an odd prime and $m \geq 1$.

§ Proof

Proof. ($\Rightarrow$) The necessity was proved earlier (Corollary ??): if a primitive root modulo n exists then n must be $1, 2, 4, p^m$, or $2p^m$.

($\Leftarrow$) For each value in the list we exhibit existence.

- $n = 1$: trivial (the residue class 0).
- $n = 2$: 1 is a primitive root modulo 2.
- $n = 4$: 3 is a primitive root modulo 4.
- $n = p^m$ with p odd prime: existence follows from Proposition ??.
- $n = 2p^m$ with p odd prime: existence follows from Corollary ??.

This completes the proof. □

§ Exercises

- For each $m \geq 1$, find a primitive root modulo 11^m .
- Do the same for modulus 17^m .

0.19 L19: 5.4 Index arithmetic and n-th power residues

L19: Index Arithmetic and n th-power residues

§ Setup / notation Let $m > 0$ be an integer for which a primitive root exists; fix a primitive root r modulo m . Then

$$\{r, r^2, \dots, r^{\varphi(m)}\}$$

is a complete reduced residue system modulo m .

§ Definition (Index) Let r be a primitive root modulo m . For $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$ the *index of a with respect to r* , denoted $\text{ind}_r(a)$, is the least positive integer n such that

$$r^n \equiv a \pmod{m}.$$

§ Remark Since $\{r^1, \dots, r^{\varphi(m)}\}$ is a reduced-residue system, $\text{ind}_r(a)$ exists and satisfies $1 \leq \text{ind}_r(a) \leq \varphi(m)$.

§ Example (indices modulo 7) Take $r = 3$, a primitive root modulo 7. Compute

$$3^1 \equiv 3, 3^2 \equiv 2, 3^3 \equiv 6, 3^4 \equiv 4, 3^5 \equiv 5, 3^6 \equiv 1 \pmod{7},$$

hence

$$\text{ind}_3(3) = 1, \text{ind}_3(2) = 2, \text{ind}_3(6) = 3, \text{ind}_3(4) = 4, \text{ind}_3(5) = 5, \text{ind}_3(1) = 6.$$

§ Proposition 5.20 (Properties of indices) Let r be a primitive root modulo m and let $a, b \in \mathbb{Z}$ with $\gcd(a, m) = \gcd(b, m) = 1$. Then

$$[(a)]_{\text{ind}_r(1)} = \varphi(m) \text{ (equivalently } \text{ind}_r(1) \equiv 0 \pmod{\varphi(m)}). \text{ind}_r(r) = 1. \text{ind}_r(ab) \equiv \text{ind}_r(a) + \text{ind}_r(b) \pmod{\varphi(m)}.$$

For $n \in \mathbb{Z}_{>0}$: $\text{ind}_r(a^n) \equiv n \text{ind}_r(a) \pmod{\varphi(m)}$.

3. § Proof

Proof. (a) and (b) are immediate from definitions: $r^{\varphi(m)} \equiv 1 \pmod{m}$ and $r^1 \equiv r \pmod{m}$. (c) If $r^{\text{ind}_r(a)} \equiv a \pmod{m}$ and $r^{\text{ind}_r(b)} \equiv b \pmod{m}$ then multiplying gives

$$r^{\text{ind}_r(a) + \text{ind}_r(b)} \equiv ab \pmod{m}.$$

By definition $r^{\text{ind}_r(ab)} \equiv ab \pmod{m}$, so by Proposition ?? (periodicity mod $\varphi(m)$) we obtain the congruence in (c). (d) follows similarly: $r^{\text{ind}_r(a^n)} \equiv a^n \equiv (r^{\text{ind}_r(a)})^n \equiv r^{n \text{ind}_r(a)} \pmod{m}$, hence the stated congruence. $\square$

§ Solving multiplicative congruences by indices Suppose $\gcd(a, m) = \gcd(b, m) = 1$ and $n > 0$. Consider the congruence

$$ax^n \equiv b \pmod{m}. \tag{0.4}$$

Taking indices (base r) converts (??) into the linear congruence

$$\text{ind}_r(a) + n \text{ind}_r(x) \equiv \text{ind}_r(b) \pmod{\varphi(m)},$$

or equivalently

$$n \text{ind}_r(x) \equiv \text{ind}_r(b) - \text{ind}_r(a) \pmod{\varphi(m)}. \tag{0.5}$$

Thus solving (??) reduces to solving the linear congruence (??) for $\text{ind}_r(x)$; standard linear congruence theory applies.

§ Example (solve $6x^4 \equiv 3 \pmod{7}$ via indices) Using the primitive root $r = 3$ modulo 7, we have $\text{ind}_3(6) = 3$ and $\text{ind}_3(3) = 1$. The congruence becomes

$$4 \text{ind}_3(x) \equiv \text{ind}_3(3) - \text{ind}_3(6) \equiv 1 - 3 \equiv -2 \equiv 4 \pmod{6}.$$

So $4 \text{ind}_3(x) \equiv 4 \pmod{6}$. Divide by $\gcd(4, 6) = 2$ to obtain

$$2 \text{ind}_3(x) \equiv 2 \pmod{3} \implies \text{ind}_3(x) \equiv 1 \pmod{3}.$$

Thus $\text{ind}_3(x) \in \{1, 4\} \pmod{6}$, giving

$$x \equiv 3^1 \equiv 3 \pmod{7} \quad \text{or} \quad x \equiv 3^4 \equiv 4 \pmod{7}.$$

Hence the two incongruent solutions are $x \equiv 3, 4 \pmod{7}$.

§ Definition (n th-power residue) Let $a, m, n \in \mathbb{Z}$ with $m > 0, n > 0$ and $\gcd(a, m) = 1$. We say that a is an n th-power residue modulo m if the congruence $x^n \equiv a \pmod{m}$ has a solution.

§ Examples

- 6 is a 3rd-power residue modulo 7 because $3^3 \equiv 6 \pmod{7}$.
- 3 is a 4th-power residue modulo 13 because $2^4 \equiv 16 \equiv 3 \pmod{13}$.
- 3 is *not* a 4th-power residue modulo 7 (one checks there is no solution x with $x^4 \equiv 3 \pmod{7}$).

§ Theorem 5.21 (Criterion for n th-power residues) Let $m > 0$ have a primitive root r . Fix $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$ and $n > 0$. Let $d = \gcd(n, \varphi(m))$. Then a is an n th-power residue modulo m iff

$$a^{\varphi(m)/d} \equiv 1 \pmod{m}.$$

Moreover, if a is an n th-power residue then the congruence $x^n \equiv a \pmod{m}$ has exactly d incongruent solutions modulo m .

§ Proof

Proof. Write the congruence $x^n \equiv a \pmod{m}$ in index form (base r):

$$n \text{ind}_r(x) \equiv \text{ind}_r(a) \pmod{\varphi(m)}.$$

This linear congruence in $\text{ind}_r(x)$ has a solution iff $d = \gcd(n, \varphi(m))$ divides the right-hand side $\text{ind}_r(a)$. By elementary number theory (Theorem on solvability of linear congruences), this divisibility condition is the solvability criterion.

The divisibility condition $d \mid \text{ind}_r(a)$ is equivalent to

$$\frac{\varphi(m)}{d} \text{ind}_r(a) \equiv 0 \pmod{\varphi(m)},$$

which in turn is equivalent to

$$r^{\frac{\varphi(m)}{d} \text{ind}_r(a)} \equiv 1 \pmod{m}.$$

But $r^{\text{ind}_r(a)} \equiv a$, so the displayed congruence is exactly $a^{\varphi(m)/d} \equiv 1 \pmod{m}$. Thus the solvability criterion is equivalent to the condition stated.

When solvable, the linear congruence $ny \equiv \text{ind}_r(a) \pmod{\varphi(m)}$ has exactly d incongruent solutions $y \pmod{\varphi(m)}$, hence yields exactly d incongruent values of $x = r^y$ modulo m . This proves the counting statement. $\square$

§ Corollary 5.22 (Quadratic residues via Euler) Let p be an odd prime and $a \in \mathbb{Z}$ with $p \nmid a$. Then a is a quadratic residue modulo p iff

$$a^{(p-1)/2} \equiv 1 \pmod{p}.$$

Moreover, if a is a quadratic residue then $x^2 \equiv a \pmod{p}$ has exactly two incongruent solutions modulo p .

§ Proof

Proof. Apply Theorem ?? with $m = p$ and $n = 2$. Then $d = \gcd(2, p-1) = 2$, and the criterion becomes $a^{(p-1)/2} \equiv 1 \pmod{p}$. When solvable there are $d = 2$ solutions modulo p . $\square$

§ Examples and exercises

- Example: Take $a = 6, m = 7, n = 3$. Here $\varphi(7) = 6$ and $d = \gcd(3, 6) = 3$. Check $6^{6/3} = 6^2 \equiv 36 \equiv 1 \pmod{7}$, so 6 is a 3rd-power residue modulo 7. By Theorem ?? the congruence $x^3 \equiv 6 \pmod{7}$ has exactly 3 incongruent solutions (indeed they are $x \equiv 3, 3 \cdot 3, 3^3$ etc., in the corresponding index classes).
- Exercise: Is 3 a 4th-power residue modulo 13? If so, how many incongruent solutions does $x^4 \equiv 3 \pmod{13}$ have? (*Hint:* $\varphi(13) = 12, d = \gcd(4, 12) = 4$; check $3^{12/4} = 3^3 \pmod{13}$.)
- Example: 15th-power residues modulo 9. Since $9 = 3^2$ has a primitive root and $\varphi(9) = 6$, we get $d = \gcd(15, 6) = 3$. The criterion is $a^{6/3} = a^2 \equiv 1 \pmod{9}$, whose coprime solutions are $a \equiv 1, 8 \pmod{9}$. Thus the 15th-power residues mod 9 are 1, 8. (Equivalently, the 15-th powers coincide with 3rd powers mod 9 because $15 \equiv 3 \pmod{6}$.)